

1 Carryover representation and inference for biomarker-treatment
2 interaction tests in aggregated N-of-1 trials under covariance
3 moderation: a factorial simulation comparison of nine analysis
4 specifications

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41 Abstract

42 **Background.** Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to vali-
43 date predictive biomarkers. Under covariance moderation, a multivariate-normal differential-
44 correlation architecture in which the biomarker-treatment interaction lives in a treatment-
45 state-dependent correlation, a companion study found that carryover costs up to roughly 31%
46 of power in relative terms. The analyst must still decide how to represent residual exposure
47 in the linear predictor, and it is not established which representation is most robust to the
48 ordinarily unknown pharmacokinetic decay shape.

49 **Methods.** We compare nine analysis-model specifications for the drug-exposure regressor, in
50 two tiers. The Tier 1 comparison, our primary evidence base, covers three specifications across
51 a full factorial grid. Unadjusted is the binary on-drug indicator, ignoring carryover altogether.
52 Lag-adjusted augments that indicator with a lagged just-off-drug term estimating carryover
53 non-parametrically, the classical crossover device. Exposure-weighted is a continuous indica-
54 tor committing to a parametric decay and half-life, the current package default. We embed
55 these three in an ADEMP-structured factorial simulation¹ crossing two data-generating decay
56 forms (exponential, Weibull) with three trial designs, two sample sizes, and three interaction
57 strengths (216 cells, 500 replicates each), plus sensitivity blocks and a cluster-robust recal-
58 ibration. All three are fitted to the same simulated datasets, so every contrast is paired, and
59 each performance measure is reported with its Monte Carlo standard error (MCSE). A Tier
60 2 extension adds six further specifications at a single reference cell: an AIC-selected half-
61 life variant, a paired-difference regression with no repeated-measures model at all, and CR2
62 cluster-robust variants of four of the above. This second tier is a more targeted comparison,
63 not a full grid.

64 **Results.** Three findings stand out. First, the lagged-treatment term buys nothing. Lag-
65 adjusted and Unadjusted are statistically indistinguishable across all 216 Tier 1 cells, differing
66 by a mean of 0.003 in power against an MCSE of 0.018, and agreeing to three decimal places
67 on bias, mean squared error, and coverage. Because the two share an estimand and differ
68 by exactly one nuisance term, this is a clean null. Second, Exposure-weighted attains the
69 highest Tier 1 power only in the two designs with a blinded-discontinuation phase (Hybrid and
70 OL+BDC), where its advantage is about 10 percentage points (0.860 vs 0.766 for Unadjusted
71 at the reference cell). Under the classical crossover (CO) design it is markedly inferior (0.488

vs 0.830). Where it leads, the lead is robust to decay-shape and half-life mis-specification and to autocorrelation strength. The model-based interaction test is mildly conservative for all three (mean null rejection 0.036 to 0.039), an artifact of the working covariance that a CR2 cluster-robust standard error removes without disturbing the ranking. Third, at the Hybrid reference cell the Tier 2 extension finds that combining the AIC-selected half-life with a CR2 standard error outperforms every other specification tested, including the CR2-corrected Exposure-weighted predictor, with Type I error held at nominal despite the post-selection risk the half-life selection step raises in principle. This result held at only that one cell, so we report it as a promising direction rather than a validated recommendation.

Conclusions. Four points warrant emphasis. First, under the covariance-moderation architecture we recommend the exposure-weighted predictor, reported with a cluster-robust standard error, for the Hybrid and OL+BDC designs. Under a crossover design it should not be used, and the unadjusted binary indicator is materially better. Second, the classical lagged-treatment remedy cannot be recommended in any design examined here, since it never improved on ignoring carryover altogether. We trace this to its entering the model as a nuisance main effect, which leaves the attenuation of the interaction untouched. Third, across all conditions anticipated dropout governs power far more than the choice among specifications, and should dominate design effort accordingly. Fourth, the AIC-selected half-life specification with a CR2 correction merits further validation as a candidate default, having outperformed every other specification at the one cell where all nine were compared directly, but it has not yet been tested against the same robustness battery the Tier 1 three have. This manuscript restricts the grid to the covariance-moderation architecture and to the exponential and Weibull decay forms. The mean-moderation architecture and linear decay are evaluated in the archived companion drafts `archive/report.Rmd` and `archive/report-slim.Rmd`.

1 Introduction

In a single N-of-1 trial, one patient is crossed repeatedly between active treatment and a comparator, and the within-patient contrast between on-drug and off-drug periods estimates that patient's individual treatment effect. An aggregated N-of-1 trial pools many such series into a mixed model, trading some within-patient resolution for population-level inference at sample sizes considerably smaller than a parallel-group trial would require². The estimand is the biomarker-treatment interaction, whether a baseline covariate B_i modifies the treatment effect. We examine three trial designs that differ in how on-drug and off-drug occasions are arranged, a classical crossover (CO), an open-label design followed by blinded discontinuation (OL+BDC), and a Hybrid N-of-1 design, each defined in Section 2.2.

The inferential complication specific to this setting is carryover. A drug's pharmacological effect does not switch off abruptly at discontinuation but decays over days or weeks, so nominally off-drug observations still carry residual exposure. An exposure regressor that ignores this is a mis-measured version of what the patient actually received, and the damage is twofold. The interaction contrast is diluted, because occasions still carrying a substantial fraction of the drug effect are coded identically to occasions carrying none, biasing the biomarker-by-

112 exposure coefficient toward zero as any covariate measured with error would. And those same
113 occasions are heterogeneous in a way the model does not represent, so the unexplained varia-
114 tion inflates the standard error of the coefficient that has just been attenuated. The test loses
115 on both counts, and power falls accordingly.

116 This is a loss of power, not of validity. Under the null, $c_{bm} = 0$, no biomarker-response
117 association exists at any level of residual exposure, so a mis-coded regressor has no interaction
118 to distort and test size is unaffected by exposure coding. The mild conservatism observed
119 in Section 3.6 has a separate origin in the working covariance and applies equally across
120 specifications. Carryover mis-specification is therefore a question of efficiency, and we frame
121 the study in terms of recovered power.

122 Three answers are available for how to represent residual exposure in the linear predictor,
123 corresponding to three positions an analyst can take toward the unknown decay. The analyst
124 may deny it, coding exposure strictly on or off, may estimate it, adding a nuisance term
125 whose magnitude the data determine, or may assume it, committing to a decay curve and a
126 half-life. A companion study³, building on Hendrickson², shows that under the covariance-
127 moderation architecture, in which the interaction lives in the treatment-state-dependent co-
128 variance $\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, carryover erodes power directly and design-dependently.
129 At a half-life of one week, the relative loss is roughly 31% under OL+BDC but only 3 to 5%
130 under crossover and Hybrid, against 1 to 3% under an alternative architecture where the
131 biomarker moderates the mean response directly. We restrict attention here to covariance
132 moderation, where the carryover problem is largest and Hendrickson's original results were
133 obtained. That earlier work fixed the decay shape at exponential and evaluated only one
134 analysis specification, leaving open which representation of residual exposure performs best
135 when the true decay shape is not known precisely, as is ordinarily the case.

136 We compare three specifications as our primary evidence base. Unadjusted is the binary
137 on-drug indicator, ignoring carryover entirely and serving as the benchmark any remedy
138 must beat. Lag-adjusted augments it with a lagged just-off-drug term that estimates car-
139 ryover magnitude without assuming a functional form, following the classical crossover-trial
140 device of Jones and Kenward⁴. Exposure-weighted is a continuous indicator committing to
141 a parametric decay form and half-life, the current pmsimstats-ng default. Unadjusted and
142 Lag-adjusted estimate the same coefficient and are fitted to the same simulated datasets, dif-
143 fering by exactly one nuisance term, so their contrast isolates that term's contribution alone.
144 Exposure-weighted differs on a second axis as well, since it changes the regressor carrying
145 the interaction. We cross all three against exponential and Weibull data-generating decay
146 forms and ask whether one can be recommended outright, or whether the choice depends
147 on trial design, following the ADEMP reporting framework of Morris, White, and Crowther¹
148 throughout. Six further specifications, an AIC-selected half-life variant, a paired-difference
149 regression, and CR2 cluster-robust variants of four base specifications, are added later at a
150 single reference cell (Section 2.5) once the results from these first three raise specific new
151 questions the Tier 1 grid alone cannot answer. Section 2 describes the simulation design in
152 ADEMP order, Section 3 gives the results, and Section 4 takes up the practical implications.

153 2 Simulation design

154 The study simulates complete aggregated N-of-1 trials and analyzes each one under every
155 specification. A single replicate proceeds as follows. Patients are allocated across the random-
156 ization paths of the chosen trial design (Section 2.2). For each patient we draw a pre-treatment
157 biomarker together with the three latent response components, namely the pharmacological
158 response, the natural-history trend and the placebo-belief component, jointly across all eight
159 measurement occasions from a multivariate normal distribution whose means follow modified
160 Gompertz trajectories and whose covariance carries within-component AR(1) autocorrelation,
161 cross-component correlations, and the treatment-state-dependent biomarker-response correla-
162 tion that encodes the interaction (Section 2.3). The observed symptom score at each occasion
163 is the patient’s baseline severity less the sum of the three components, so that a beneficial
164 effect lowers the score. Carryover enters through a decay function of time since discontinua-
165 tion, which keeps part of the pharmacological effect present at nominally off-drug occasions.
166 All three analysis specifications (Section 2.4) are then fitted to that same simulated dataset,
167 and we record the interaction coefficient, its standard error and its p -value from each. This
168 is repeated 500 times for every parameter combination in the grid of Section 2.7.

169 Fitting all specifications to a common dataset is deliberate. It makes every contrast among
170 them paired, so that differences are estimated with substantially less Monte Carlo variance
171 than independent draws would give. Throughout, N denotes the **total** number of patients in
172 a trial, allocated as evenly as possible across the design’s randomization paths, so a design
173 with P paths assigns about N/P patients to each. At $N = 70$ this is 35 per path under the
174 two-path designs and 17 or 18 per path under the four-path Hybrid design.

175 2.1 Aims

176 The aim is to quantify, under covariance moderation, the cost of mis-specifying the carryover
177 representation in the analysis model, and to determine which of the three candidate strategies
178 performs better across the grid of data-generating decay forms and trial designs. A second
179 aim, addressed in the Tier 2 sensitivity blocks of Section 3.7, is to check whether the result-
180 ing ranking survives perturbation of within-subject autocorrelation, dropout rate, half-life
181 accuracy, and interaction size.

182 2.2 Trial designs

183 The three designs differ in how on-drug and off-drug occasions are arranged, and so in how
184 much post-discontinuation information each generates. All three schedule eight measurement
185 occasions per patient across randomization paths, each a fixed on-drug/off-drug schedule,
186 and each design carries an expectancy weight (one during open-label periods, one half during
187 blinded periods) scaling the placebo-belief component.

188 **Crossover (CO).** A traditional two-period within-patient crossover, blinded throughout,
189 occasions spaced 2.5 weeks apart from 2.5 to 20 weeks. One path is active for the first
190 four occasions then comparator, and the other reverses that order and so never discontinues,
191 contributing no post-discontinuation observations.

192 **Open-label followed by blinded discontinuation (OL+BDC).** Open-label titration at
 193 4, 8, 12, and 16 weeks, then weekly follow-up to week 20 with a blinded late switch to placebo.
 194 The two paths discontinue after week 17 or 18, so only two or three occasions follow.

195 **Hybrid.** The Hendrickson design and this program's reference. Open-label titration at
 196 weeks 4 and 8 is followed by blinded discontinuation weekly from week 9 to 12, then a brief
 197 two-occasion crossover at weeks 16 and 20. The four paths vary discontinuation timing and
 198 crossover-arm order. It is the richest of the three in on-off transitions and the only one allowing
 199 more than one discontinuation per patient.

200 One consequence bears directly on the results. The first post-discontinuation occasion falls
 201 one week after switching under OL+BDC and Hybrid, but 2.5 weeks under CO. At the half-
 202 lives examined here, roughly half the pharmacological effect therefore persists into the first
 203 off-drug occasion under the two non-crossover designs, against less than a fifth under CO, so
 204 the designs differ substantially in how much carryover contamination they can accumulate at
 205 all (Section 4.2).

206 2.3 Data-generating mechanism

207 Under covariance moderation, the biomarker and the biological response are drawn jointly
 208 from a multivariate normal distribution with a treatment-state-dependent correlation,
 209 $\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form and t_{sd} the time since drug discon-
 210 tinuation. Mean moderation and the combined dual-channel architecture fall outside this
 211 manuscript's scope. The full cross-architecture comparison is in the archived full-scope drafts
 212 (`archive/report.Rmd`, `archive/report-slim.Rmd`). The decay function $\phi : \mathbb{R}_+ \rightarrow [0, 1]$
 213 maps t_{sd} to a residual-effect multiplier, and we evaluate two forms here, each matched to a
 214 common half-life $t_{1/2}$. A third, linear decay, appears in the full-scope manuscripts but not
 215 here.

216 2.3.1 Exponential decay

$$217 \quad \phi_{\text{exp}}(t_{sd}) = \exp(-\lambda t_{sd}), \quad \lambda = \frac{\ln 2}{t_{1/2}}$$

218 This is the decay form used throughout the companion manuscript and implemented in
 219 `functions.R` under `implementations/tidyverse/`. It corresponds to classical first-order
 220 elimination kinetics, the default pharmacokinetic assumption, and serves as the reference
 221 against which Weibull is compared.

222 2.3.2 Weibull decay

$$223 \quad \phi_{\text{weib}}(t_{sd}) = \exp\left(-(\lambda_w t_{sd})^k\right), \quad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

224 A shape parameter k governs whether the elimination tail is heavier than exponential ($k <$
 225 1, slower elimination) or lighter ($k > 1$, faster clearance), recovering the exponential form
 226 exactly at $k = 1$. We report $k = 0.5$ (a markedly prolonged low-concentration tail) and

227 $k = 2.0$ (markedly faster clearance once concentration falls below some threshold), a wider
 228 sensitivity range than an earlier version of this grid used ($k = 0.7, 1.5$). $k = 1.0$ is not a third
 229 Weibull condition: it is numerically indistinguishable from Exponential and so is not reported
 230 separately, giving three decay-shape levels in the grid rather than four.

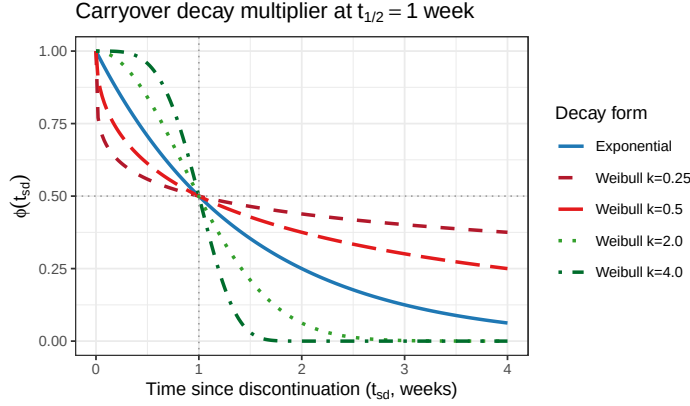


Figure 1: The carryover decay multiplier $\phi(t_{sd})$ at the reference half-life ($t_{1/2} = 1.0$ week) for exponential decay and the two Weibull shapes. Both dotted reference lines mark the defining half-life point ($\phi = 0.5$ at $t_{sd} = t_{1/2}$), which all three curves pass through by construction; the shape parameter governs only how quickly ϕ falls before that point and how slowly it decays after.

231 Figure 1 makes the shape difference concrete. The $k = 0.5$ curve has a much heavier tail, and
 232 ϕ stays well above the exponential curve for every $t_{sd} > t_{1/2}$. In practical terms, the analysis-
 233 side predictor picks up substantially more residual carryover during a prolonged washout
 234 than the half-life alone would suggest. The $k = 2.0$ curve has a much lighter tail and clears
 235 sharply faster once concentration falls below the half-maximum threshold. All three curves
 236 pass through the same half-life-defining point ($\phi = 0.5$ at $t_{sd} = 1.0$), so the three forms
 237 remain comparable at a fixed nominal half-life even though their shapes diverge substantially
 238 on either side of it.

239 2.4 Estimand and analysis specifications

240 The interaction estimand is the on-drug biomarker-response correlation c_{bm} , a dimen-
 241 sionless quantity. Bias and empirical SE are reported on a calibrated effect-size scale
 242 (the expected difference in on-drug response per SD of the biomarker at $t_{1/2} = 0$, per
 243 docs/19-mean-moderation-implementation-notes.tex). Power and Type I error are
 244 on the native test-statistic scale, with the null estimand $c_{bm} = 0$. The analysis model is
 245 `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~`
 246 `t | ptID))` for the Tier 1 three, which differ only in how X is constructed and whether a
 247 nuisance term is added. Unadjusted and Lag-adjusted both test the coefficient on $bm:D_{it}$, so
 248 every performance measure is comparable between them. Exposure-weighted tests $bm:D_{bc,it}$,
 249 a different quantity. Power and Type I error remain comparable across all three, since they
 250 are properties of the test, but bias, MSE, and coverage are not comparable across that divide
 251 (Section 3.1). Nine specifications are evaluated in total across this manuscript: the Tier 1

Table 1: All nine analysis specifications evaluated in this manuscript. G1-G3 are the Tier 1 headline three (Section 2.4); G4-G5 are Tier 2 additions replacing the retired E4/E5/E6 (Blocks S7-S9, Section 3.7); G6-G9 are CR2 cluster-robust variants of G1, G2, G4, and G3 respectively (Block S10). G10 (G5+CR2) does not apply and is omitted: G5 collapses to one row per patient before fitting, leaving no repeated-measures cluster structure for CR2 to correct.

Code	Stored	Name	Formula	Target coefficient
G1	E1	Unadjusted	Sx bm + t + Db + bm:Db	bm:Db
G2	E3	Lag-adjusted	Sx bm + t + Db + bm:Db + L	bm:Db
G3	E2	Exposure-weighted	Sx bm + t + Dbc + bm:Dbc	bm:Dbc
G4	E7	AIC-selected t1/2	G3 formula, t1/2 by AIC over {0.25,0.5,1.0,2.0}	bm:Dbc
G5	E9	Paired-difference	diff bm (patient-level OLS)	bm slope
G6	E1cr2	Unadjusted + CR2	G1 formula, CR2 SE	bm:Db
G7	E3cr2	Lag-adjusted + CR2	G2 formula, CR2 SE	bm:Db
G8	E2cr2	Exposure-weighted + CR2	G3 formula, CR2 SE	bm:Dbc
G9	E7cr2	AIC-selected t1/2 + CR2	G4 formula, CR2 SE	bm:Dbc

three below, plus six Tier 2 additions introduced in the ‘Choice of analysis specifications’ section that follows and tested in Blocks S7-S10 (Section 3.7). Internally, and in the code and data files described in the Reproducibility section, each specification is stored under an E-prefixed code (E1, E2, ...); this manuscript’s prose refers to all nine uniformly by the compact G1-G9 code defined in `spec-labels.R`, or by name, per Table 1 below.

2.4.1 Unadjusted (G1): binary on-drug indicator

Here $X = D_{it}$, the binary on-drug indicator (one on drug, zero off). Carryover is not represented at all. An off-drug occasion still carrying most of the pharmacological effect is therefore coded identically to one carrying none. This is the default an analyst obtains without considering carryover, and we treat it as the benchmark the other specifications must justify their additional structure against. It consumes no half-life, so it is invariant to the analyst’s pharmacokinetic assumptions.

2.4.2 Lag-adjusted (G2): binary indicator plus estimated carryover term

We retain $X = D_{it}$ and augment the model with a lagged-treatment indicator, $L_{it} = D_{i,t-1}(1 - D_{it})$, marking off-drug timepoints that immediately follow an on-drug one. This is the classical crossover-trial device of Jones and Kenward⁴. Carryover magnitude is estimated from the data rather than assumed, so, like Unadjusted, this specification is invariant to the assumed half-life. L_{it} enters only as a nuisance main effect; there is no $\text{bm}:L_{it}$ product, so the interaction actually tested remains $\text{bm}:D_{it}$. Lag-adjusted can absorb mean-level carryover, but it cannot repair the interaction’s attenuation (Section 1). We think of it as Unadjusted with a carryover nuisance control added, not as a fundamentally different model of the interaction (Section 4.1).

2.4.3 Exposure-weighted (G3): continuous decayed predictor

Here $X = \text{Dbc}_{it}$, a continuous exposure-decayed predictor matched to the analyst’s assumed decay form and half-life; this is the current pmsimstats-ng default. At on-drug timepoints

276 $D_{bc,it} = 1$. Off drug it decays as $D_{bc,it} = (1/2)^{t_{sd,it}/\hat{t}_{1/2}}$, losing half its value every $\hat{t}_{1/2}$ weeks
 277 after discontinuation. When the assumed decay form matches the true one, the predictor is
 278 well matched. When it does not, it is mis-specified, and we probe the consequence directly
 279 in Block S2. The half-life assumption enters only through the predictor values handed to
 280 `lme`, not through the model formula, so Exposure-weighted is in effect a different model for
 281 each assumed half-life. It is the only one of the three original specifications that consumes
 282 a half-life at all, and this is what the S2 sensitivity contrast (Section 3.7) is designed to
 283 probe. A limiting relationship connects Exposure-weighted back to Unadjusted. As $\hat{t}_{1/2} \rightarrow 0$,
 284 $D_{bc,it} \rightarrow D_{it}$, so Unadjusted is exactly the boundary member of the Exposure-weighted family
 285 at zero assumed half-life. The S2 sweep is therefore a traverse anchored at Unadjusted, not
 286 an independent check.

287 2.4.4 AIC-selected half-life (G4)

288 This specification uses Exposure-weighted's formula, $X = D_{bc,it}$, but chooses $\hat{t}_{1/2}$ from the
 289 data rather than assuming it. Writing $M(t_{1/2})$ for the `lme` fit of Section 2.4 with $D_{bc,it}$
 290 constructed at a given candidate half-life, the selected half-life is

$$291 \quad \hat{t}_{1/2} = \arg \min_{t_{1/2} \in \mathcal{T}} \text{AIC}(M(t_{1/2})), \quad \mathcal{T} = \{0.25, 0.5, 1.0, 2.0\} \text{ weeks},$$

292 with $\text{AIC}(M) = -2 \log L(M) + 2p$, $L(M)$ the fitted model's likelihood and p its number of
 293 estimated parameters⁵. Reported power, the interaction coefficient, and its standard error
 294 are then read from $M(\hat{t}_{1/2})$ alone, the single winning fit, not pooled or averaged across $\mathcal{T}$.
 295 This mirrors `characterize_carryover()` (`R/carryover_analysis.R`), the package's own
 296 real-data half-life-selection routine. Like Exposure-weighted, it targets `bm:Dbc,it`. Unlike
 297 Exposure-weighted, it requires no half-life from the analyst at all. The cost is a post-selection
 298 inference risk⁶: $M(\hat{t}_{1/2})$'s reported standard error and p -value treat $\hat{t}_{1/2}$ as though it were fixed
 299 in advance, when the same data in fact chose it, so the nominal inference does not formally
 300 account for the selection step. Block S11 (Section 3.7) checks whether this risk manifests as
 301 Type I error inflation in practice.

302 2.4.5 Paired-difference (G5)

303 This is a two-stage, correlation-structure-agnostic alternative, rather than an elaboration of
 304 the mixed-model family above. For each patient i with $n_i^{\text{on}} > 0$ on-drug and $n_i^{\text{off}} > 0$ off-drug
 305 occasions, define the patient's own on-drug and off-drug means,

$$306 \quad \bar{S}_{x,i}^{\text{on}} = \frac{1}{n_i^{\text{on}}} \sum_{t: D_{it}=1} S_{x,it}, \quad \bar{S}_{x,i}^{\text{off}} = \frac{1}{n_i^{\text{off}}} \sum_{t: D_{it}=0} S_{x,it},$$

307 and their difference $\Delta_i = \bar{S}_{x,i}^{\text{on}} - \bar{S}_{x,i}^{\text{off}}$. The second stage regresses this single per-patient value
 308 on the (centered) biomarker across the $N' \leq N$ patients contributing a difference,

$$\Delta_i = \beta_0 + \beta_1 bm_i + \varepsilon_i, \quad i = 1, \dots, N',$$

by plain OLS, with the interaction test on β_1 . There is no repeated-measures model, no carryover representation, and no correlation structure of any kind: Δ_i collapses each patient's eight occasions to one number before bm ever enters the model. We view it as a direct biomarker-interaction extension of the paired t-test that Senn, Julious and Araujo⁷ found to outperform carryover-adjusted mixed models for ordinary N-of-1 treatment-effect estimation. Patients contributing only on-drug or only off-drug observations (for instance, CO's never-discontinuing path) have $n_i^{\text{on}} = 0$ or $n_i^{\text{off}} = 0$, cannot form Δ_i , and are dropped, so $N' < N$ whenever such a path exists in the design.

2.4.6 CR2 cluster-robust variants (G6-G9)

Section 3.6 (Block S6) shows that the model-based interaction test is mildly conservative, and that a CR2 bias-reduced cluster-robust standard error restores nominal size without disturbing the ranking among G1-G3. We use the estimator of Bell and McCaffrey⁸, via `clubSandwich`, paired with Satterthwaite degrees of freedom. That correction, `lme+CR2` in Section 2.7, was applied only to G3 there. Block S10 (Section 3.7) extends it to G1, G2, and G4 as well, replacing only the standard error, reference distribution, and p -value on each specification's existing point estimate: G6 (G1+CR2), G7 (G2+CR2), G9 (G4+CR2). G8 (G3+CR2) is the same specification already introduced in Section 2.7 and Block S6, and we carry it into the nine-way G1-G9 comparison at Block S10 without rerunning it. A tenth combination, G5+CR2, does not apply. G5 already collapses to one row per patient before fitting, so there is no repeated-measures cluster structure left for CR2 to correct.

2.5 Choice of analysis specifications

Three considerations bounded which specifications were tested, rather than surveying the wider space of models used in the N-of-1 and crossover-trial literature.

The Tier 1 three (Unadjusted, Lag-adjusted, Exposure-weighted, above) were chosen because each represents a distinct, established position an analyst can take toward an unmeasured or assumed pharmacokinetic decay, not because they exhaust that space. Unadjusted is the default any analyst obtains without considering carryover at all, and so is the necessary benchmark for every alternative. Lag-adjusted follows Jones and Kenward's⁴ classical crossover-trial device, the shape-free remedy the pre-mixed-model crossover literature has long recommended. Exposure-weighted is the `pmsimstats-ng` package's own established default, so it is the specification any user of the package is actually running today, and the manuscript's practical stake in getting its behavior right is correspondingly higher than for the other two.

This left a wide range of alternative model classes untested by design: Bayesian hierarchical N-of-1 models; GEE and other marginal models, compared systematically rather than as a robustness check on one specification's standard error (Section 2.7); randomization and permutation tests; functional or spline-based flexible decay curves; state-space or Kalman-filter

346 treatments of drug concentration; and two-stage ‘series of N-of-1 trials’ meta-analytic combi-
347 nation. Each is established in the N-of-1 or crossover-trial literature, and none is evaluated
348 here. This is a scoping decision, not an oversight: the manuscript’s aim is to compare carry-
349 over *representations* within a single, fixed conditional mixed-model framework, matching
350 Hendrickson’s² original analysis model, not to conduct a comprehensive methods bake-off. A
351 wider survey belongs to future work (Section 4.6).

352 Blocks S7 through S9 (Section 3.7) revisit this scoping decision once, for a specific reason.
353 Section 4.1 shows that no carryover term tested so far can repair the interaction’s attenuation,
354 no matter how it enters, as long as it enters only as a nuisance main effect. The natural next
355 model interacts the carryover term with the biomarker directly. An earlier version of Block
356 S7 tested two such specifications, E5 (a lagged term crossed with `bm`) and E6 (a continuous
357 elapsed-time term crossed with `bm`). Both underperformed Unadjusted at every cell examined
358 across three blocks, two trial designs, two architectures, and three carryover half-lives (Sections
359 3.6, 4.5), traced to a structural cause: the data-generating process examined throughout this
360 manuscript has a single true decay parameter, so crossing a second carryover term with the
361 biomarker adds a free parameter with no counterpart in the truth being estimated, and two
362 collinear regressors chasing one true degree of freedom lose power to a correctly specified
363 single-parameter model almost by construction.

364 We expect a further variant of the same interacted-term family to fail on that same structural
365 argument, so rather than test one we replaced E5/E6 with two specifications that answer
366 different open questions about the same underlying problem, the analyst’s unknown true
367 decay half-life. G4 asks whether estimating the half-life from the data recovers the power
368 a correctly assumed half-life would give. It selects the half-life by AIC over a candidate
369 grid, reusing the logic of `characterize_carryover()` (R/`carryover_analysis.R`), already
370 used for real-data analysis in this package. This addresses Block S2’s question from the
371 opposite direction: instead of asking how much a wrong assumption costs, we ask whether
372 estimation avoids needing the assumption at all. G5 asks a more basic question raised directly
373 by the wider literature: is a conditional mixed model even necessary? Senn, Julious and
374 Araujo⁷ compared four analysis strategies for ordinary (no biomarker interaction) N-of-1 trials.
375 They found that a simple paired t-test on within-patient treatment differences achieved near-
376 nominal Type I error, higher power, and comparable bias and MSE relative to a carryover-
377 adjusted mixed model, and recommended the mixed model only when carryover is specifically
378 suspected. G5 is a direct biomarker-interaction extension of their preferred method: a two-
379 stage, correlation-structure-agnostic regression of each patient’s own mean on-drug-minus-off-
380 drug difference on the biomarker across patients, with no repeated-measures model and no
381 carryover representation at all.

382 This remains a bounded comparison. G4 and G5 were chosen because each is implementable
383 within, or trivially adjacent to, the existing analysis pipeline (Section 2.7) and each answers
384 a specific question this manuscript’s own results raised, not because they close the wider
385 literature gap identified above. The Bayesian, GEE-as-primary-model, randomization-test,
386 and two-stage meta-analytic alternatives remain untested and are discussed as future work
387 (Section 4.6).

2.6 Factorial grid and parameter settings

The principal comparison crosses biomarker moderation, carryover half-life, trial design, and sample size against the three analysis specifications, under covariance moderation and the exponential DGP.

Parameter	Values
Biomarker moderation (c_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	B (covariance moderation) only
Replicates per cell	500 (target), 50 for development

Three specifications, three half-lives, three designs, two sample sizes, and three biomarker levels give $3 \times 3 \times 3 \times 2 \times 3 = 162$ cells. The $c_{bm} = 0$ row is the Type I error check against nominal 5%, and the $t_{1/2} = 0$ row, with no carryover, gives a best-case power reference at each combination of N , design, and c_{bm} against which the cost of nonzero carryover can be read off directly.

Decay-shape sensitivity (Figures 3 and 5) is evaluated separately, at the reference cell only ($c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks), rather than across the full grid above: exponential decay and two Weibull shapes ($k \in \{0.5, 2.0\}$, Section 2.3) crossed with the three specifications, three designs, and two sample sizes give $3 \times 3 \times 3 \times 2 = 54$ further cells (23-run-decay-shape-sensitivity.R). We restrict the decay-shape axis to this one cell because it is the axis every other result in this manuscript holds fixed at exponential; a full crossing of decay shape with the grid above would triple the simulation cost for a comparison this manuscript uses in only two figures. $k = 1.0$ is not a third Weibull condition here, or anywhere in this manuscript: it recovers the exponential form exactly (Section 2.3) and so is not a distinct decay-shape condition.

2.7 Inference and the cluster-robust correction

The interaction test described so far is model-based. The standard error of $\beta_{bm:X}$ is read from the fitted `lme` object, valid only if the assumed residual covariance is correct. That assumption is not innocuous here. The working covariance combines a patient random intercept with a `corCAR1` AR(1) residual correlation, and the companion calibration study⁹ shows this does not match the covariance the data-generating mechanism actually induces, inflating the standard error by roughly six to ten percent and depressing the rejection rate correspondingly. We quantify this with the calibration ratio

$$\kappa = \frac{\widehat{SE}}{SD(\hat{\beta}_{bm:X})} = \frac{\frac{1}{n_{sim}} \sum_{r=1}^{n_{sim}} \widehat{SE}_r}{\sqrt{\frac{1}{n_{sim}-1} \sum_{r=1}^{n_{sim}} (\hat{\beta}_{bm:X,r} - \hat{\beta}_{bm:X})^2}},$$

the mean model-reported standard error across the n_{sim} simulation replicates, divided by the empirical standard deviation of the replicate-level point estimates $\hat{\beta}_{bm:X,r}$ themselves ($\kappa = 1$ is correctly calibrated, since the model's own reported uncertainty then matches the estimator's actual sampling variability; $\kappa > 1$ means the model overstates that uncertainty, a conservative test). At the null cell κ runs 1.05 to 1.10 across specifications, with empirical Type I error 0.029 to 0.039 against nominal 0.05.

The remedy is a cluster-robust (sandwich) variance estimator, which derives the variance of $\hat{\beta}_{bm:X}$ from observed between-cluster variation rather than the assumed covariance, and so remains valid under mis-specification. Clusters are patients (eight occasions each, unique across randomization paths). Writing X_i for patient i 's rows of the fixed-effects design matrix and $\hat{e}_i = y_i - X_i\hat{\beta}$ for that patient's residual vector from the fitted model (N patients, indexed $i = 1, \dots, N$), the naive (Liang-Zeger) sandwich estimator is

$$\widehat{\text{Var}}_{\text{sandwich}}(\hat{\beta}) = \left(\sum_{i=1}^N X_i' \hat{V}_i^{-1} X_i \right)^{-1} \left(\sum_{i=1}^N X_i' \hat{V}_i^{-1} \hat{e}_i \hat{e}_i' \hat{V}_i^{-1} X_i \right) \left(\sum_{i=1}^N X_i' \hat{V}_i^{-1} X_i \right)^{-1},$$

with $\hat{V}_i$ the working covariance from the `lme` fit. This formula is exactly the source of the standard-error inflation described above, reversed: it substitutes each patient's own, directly observed $\hat{e}_i \hat{e}_i'$ for the single assumed $\hat{V}_i$ everywhere it appears in the middle term, so a mismatched working covariance no longer biases the variance estimate, at the cost of needing enough clusters for that substitution to be stable. With only $N = 35$ or 70 clusters, the naive form in $\widehat{\text{Var}}_{\text{sandwich}}$ is itself markedly downward-biased, so we use the CR2 bias-reduced estimator of Bell and McCaffrey⁸. CR2 replaces each raw residual $\hat{e}_i$ above with an adjusted residual $\hat{e}_i^{\text{CR2}} = A_i \hat{e}_i$, where A_i is a patient-specific matrix chosen so that $\widehat{\text{Var}}_{\text{sandwich}}$, recomputed with $\hat{e}_i^{\text{CR2}}$ in place of $\hat{e}_i$, is exactly unbiased under the working model (the construction of A_i is given in full in Bell and McCaffrey⁸; we do not reproduce it here). We pair this with Satterthwaite degrees of freedom (both via `clubSandwich`), replacing only the standard error, reference distribution, and p -value, leaving point estimates untouched.

We apply the correction in two places. A null-cell diagnostic ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid, $N = 70$, 3,000 replicates) establishes whether CR2 restores nominal size (Section 3.6). A five-cell block, the reference cell plus four stress cells (small N , high autocorrelation, half-life mis-specification, 30% MCAR dropout, 500 replicates each, labeled S6 in this manuscript's block numbering), then asks whether restoring size disturbs the ranking (same section). Blocks S1 through S4 are the Tier 2 sensitivity analyses of Section 3.7. Block S5, an exploratory autocorrelation-by-carryover interaction, is outside the present scope and reported only in the archived full-scope drafts.

2.8 Performance measures and computing environment

For each cell we report Power, Type I error, Bias, Empirical SE (calibrated scale), Coverage of the nominal 95% Wald CI, MSE, non-convergence rate, and the MCSE of each, following Morris et al.^{1, Tables 5-6}. We chose $n_{sim} = 500$ so the MCSE on power at $p = 0.5$ would not

exceed 0.022. No simulation runs for this manuscript. We filter the same 540-cell production grid summary used in the archived full-scope drafts down to the 216-cell slice defined by covariance moderation and the two non-linear decay forms. All three specifications are fit to the same simulated dataset within each cell (common random numbers), so every pairwise contrast, especially Lag-adjusted minus Unadjusted, carries reduced Monte Carlo variance relative to independent draws. The Tier 2 sensitivity blocks and S6 recalibration are filtered from the same shared production outputs. Exact file paths and versions are given in the Reproducibility section.

3 Results

All results reported below derive from the same 500-replicate production run used in the full-scope manuscripts, filtered to the covariance-moderation architecture and the exponential and Weibull decay forms. The Monte Carlo standard error on power does not exceed 0.022 at $p = 0.5$, and differences smaller than this should not be over-interpreted. Coverage of the nominal 95% Wald confidence interval is reported for the reference cell, and non-convergence rates are reported per cell in the tables that follow.

3.1 Headline performance table

Table 2 reports performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$), where non-convergence was zero throughout. The bias, MSE, and coverage columns are computed against Exposure-weighted's calibrated target, $\theta_{\text{true}} = -c_{bm}\sigma_{BR} = -3.6$, while Unadjusted and Lag-adjusted estimate a different coefficient, $\text{bm}:D_{it}$. Their apparent bias and under-coverage reflect that estimand mismatch, not a defect in the estimators, so those columns are comparable within the Unadjusted/Lag-adjusted pair but not across to Exposure-weighted. For the three-way comparison we rely on power and Type I error, properties of the test rather than the coefficient tested.

3.2 Tier 1 grid overview

Figures 2 through 4 give a full-grid view of the Tier 1 results as annotated heatmaps, with trial design as column facets, N as row-block facets, analysis specification as the row axis within each block, and a blue (low power) to red (high power) fill scale. Each panel fixes two of the grid's remaining dimensions and varies a third, since no single two-dimensional heatmap can display all four simultaneously. Panels A and B report all nine G1-G9 specifications (a dedicated G1-G9 simulation restricted to the exponential DGP and Architecture B, $n_{sim} = 100$, preliminary); Panel C retains the original three (G1-G3), since it is the only panel that varies DGP decay form, an axis the G1-G9 extension did not re-run.

Two features are visible at a glance in Panel A that the headline table makes only cell by cell. First, within every design by N block, G1 and G2 (Unadjusted, Lag-adjusted) track each other closely at every biomarker level, differing by at most 0.03, within Monte Carlo noise at this preliminary $n_{sim} = 100$. This matches the higher-precision result of Section 3.4, where the two are numerically identical to two decimal places. Second, the Exposure-weighted

A: Effect of trial design, analysis specification, and biomarker effect on power ($t_{1/2} = 1$)

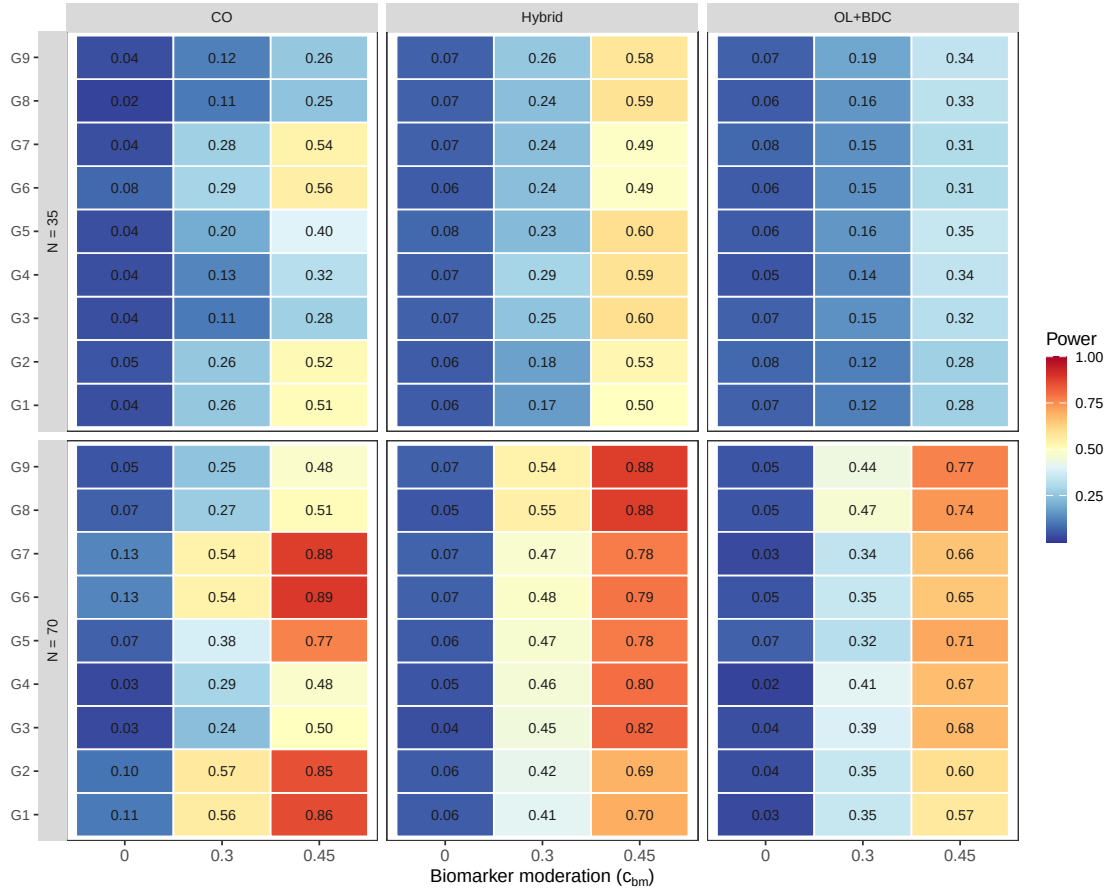


Figure 2: Tier 1 grid, biomarker-effect axis, all nine G1-G9 specifications: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1.0$ weeks, exponential DGP, Architecture B (preliminary $n_{sim} = 100$).

Table 2: Headline performance at the reference cell (exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$, the primary outcome of this manuscript. Bias is $\text{mean}(\text{estimate} - \theta_{\text{true}})$ on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration; see the estimand-mismatch caveat above for Unadjusted/Lag-adjusted). Empirical SE is the standard deviation of the point estimates across replicates, the actual sampling variability of the estimator, as distinct from whatever SE any single fit reports (the comparison the calibration ratio κ of Section 2.7 is built on). Coverage is the proportion of replicates whose nominal 95% Wald CI contains θ_{true} ; the low Unadjusted/Lag-adjusted values reflect the same estimand mismatch as Bias, not miscalibrated inference. MSE is $\text{bias}^2 + \text{Empirical SE}^2$, an overall accuracy summary. Non-conv is the fraction of replicates where ‘nlme::lme()’ failed to converge (zero throughout at this cell). MCSEs are reported per Morris et al. (2019, Table 6).

t1/2	Specification	Power (MCSE)	Bias (MCSE)	Empirical SE	Coverage (MCSE)	MSE (MCSE)	Non-conv
0.0	Unadjusted	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	Lag-adjusted	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	Exposure-weighted	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	Unadjusted	0.918 (0.012)	+0.586 (0.036)	0.807	0.938 (0.011)	0.995 (0.058)	0.000
0.5	Lag-adjusted	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
0.5	Exposure-weighted	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
1.0	Unadjusted	0.766 (0.019)	+1.093 (0.038)	0.847	0.828 (0.017)	1.911 (0.093)	0.000
1.0	Lag-adjusted	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000
1.0	Exposure-weighted	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000

family (G3, G4, and their CR2 variants G8, G9) visibly separates from G1/G2/G6/G7 only in the Hybrid and OL+BDC columns. In the CO column the whole family underperforms at every biomarker level past zero. At the reference cell ($c_{bm} = 0.45$, $t_{1/2} = 1.0$), for example, G4 reaches 0.48 against 0.86 to 0.89 for G1/G2/G6/G7. This is the design-dependence of Section 3.4, read directly off the grid now across all nine specifications rather than three. Whether that separation depends on carryover being present at all, Panel A alone cannot show, since it fixes $t_{1/2} = 1.0$ throughout.

Panel B answers that, for Hybrid and OL+BDC. At $t_{1/2} = 0$ the Exposure-weighted family’s separation from G1/G2/G6/G7 disappears in both designs (all nine specifications sit at 0.94-1.00), and only opens up as the half-life increases, confirming the separation in Panel A is specifically a carryover effect there (the $t_{1/2} = 1.0$ column is shared with Panel A’s reference cell). Under CO the picture is different: G4 and G9 are already collapsed at $t_{1/2} = 0$ (0.48 and 0.47, against 0.88-0.92 for the rest), before carryover enters the picture at all. This is the AIC mis-selection pathology of Block S9 read directly off the grid, not a carryover effect: G4’s half-life selection drifts to the largest candidate under CO’s sparse off-drug sampling regardless of the true half-life, so its CO weakness predates $t_{1/2} > 0$ rather than growing from it. Panel B also adds a ranking result Panel A alone does not show. G9 and G8 lead, in a near-tie, at the Hybrid and OL+BDC reference cells (both around 0.88 at Hybrid, 0.77 to 0.74 at OL+BDC). G5 (Paired-difference) is the strongest specification at CO that is not G1/G2 or a CR2 variant of them (0.77, versus 0.48 to 0.51 for the Exposure-weighted family there). This is consistent with G5 discarding the carryover representation that costs the Exposure-weighted family power under CO.

B: Effect of carryover half-life on power ($c_{bm} = 0.45$)

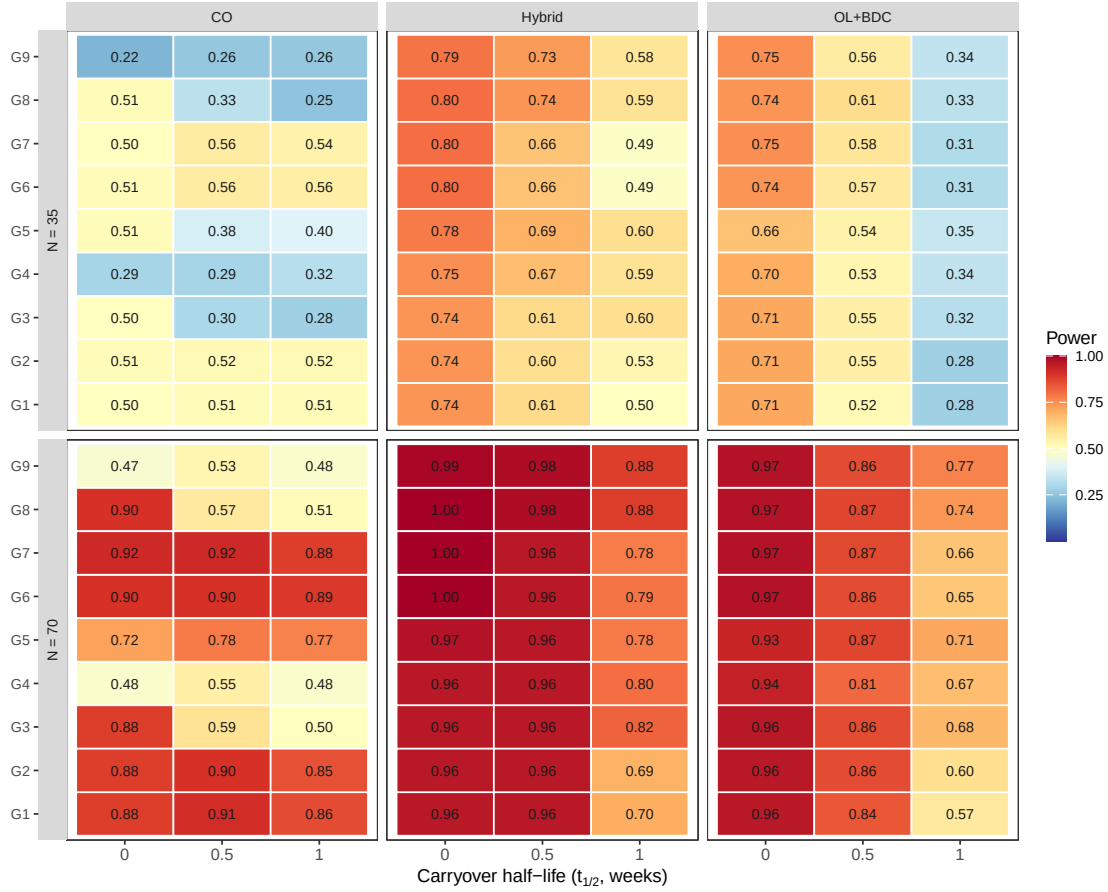


Figure 3: Tier 1 grid, carryover-axis, all nine G1-G9 specifications: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks at $c_{bm} = 0.45$, exponential DGP, Architecture B (preliminary $n_{sim} = 100$).

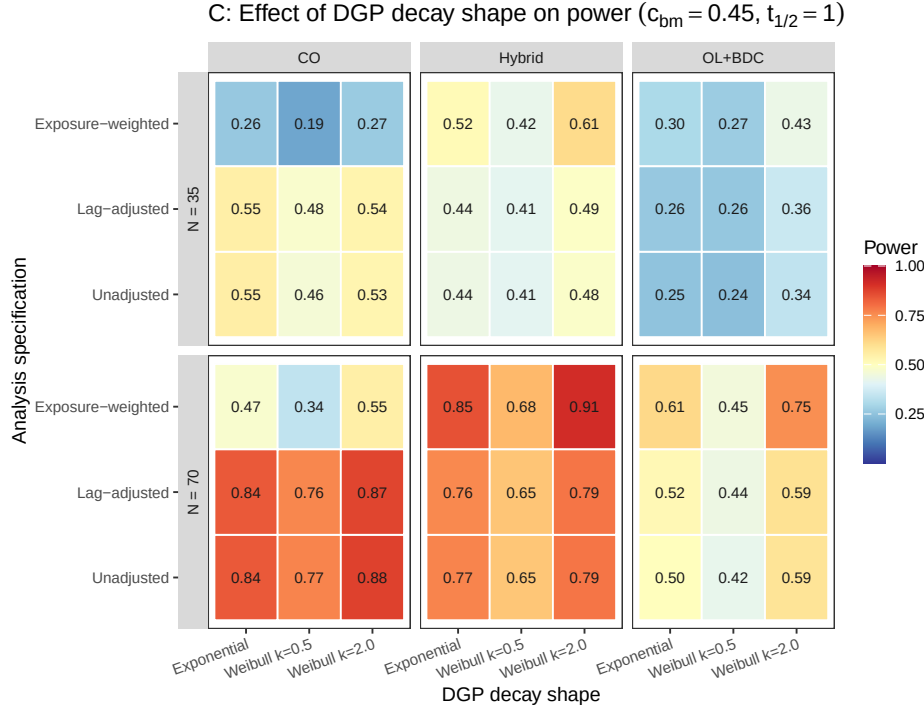


Figure 4: Tier 1 grid, decay-shape axis: power by trial design (columns), N (row blocks), and analysis specification (rows within block), across the three DGP decay-shape levels (exponential, Weibull $k \in \{0.5, 2.0\}$) at $c_{bm} = 0.45, t_{1/2} = 1.0$ weeks.

Panel C shows the Exposure-weighted row's advantage in the Hybrid and OL+BDC columns holding across every decay shape examined, including the heaviest-tailed Weibull form ($k = 0.5$), where the advantage narrows most but does not reverse. This is the decay-form robustness of Section 3.5, shown here across the full grid rather than at one cell.

3.3 Matched-decay baseline

At the matched cell (exponential DGP, exposure-weighted analysis, $N = 70, c_{bm} = 0.45$, Hybrid design), power declines from 0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks (point estimate plus or minus MCSE, $n_{sim} = 500$). All 500 replicates converged at every half-life in this cell.

We had previously read the value at $t_{1/2} = 1.0$ as agreeing with the power of 0.82 reported by Hendrickson et al. for a comparable cell². Unfortunately, we must withdraw that comparison. In the reference implementation, the analysis-side and data-generating carryover parameters are set independently. The drivers that reproduce the original figures supply a half-life to the data-generating step only, leaving the analysis-side half-life at its default of zero. At a zero half-life, the exposure-decayed predictor collapses onto the binary on-drug indicator. The reference analysis is therefore Unadjusted, not Exposure-weighted, so the appropriate comparator is our unadjusted power at the same cell, 0.766. The grids differ further on interaction strength and half-life, so 'the same cell' was approximate in any case. We have not resolved what the published figure was computed under, and make no claim of agreement

or disagreement here. Nothing else in this manuscript depends on that comparison, since the analysis specifications are compared against one another within a single simulation under common random numbers.

3.4 Power by analysis specification

Power vs carryover under matched exponential DGP

Covariance-moderation architecture, $N = 70$, $c_{bm} = 0.45$, 500 reps/cell

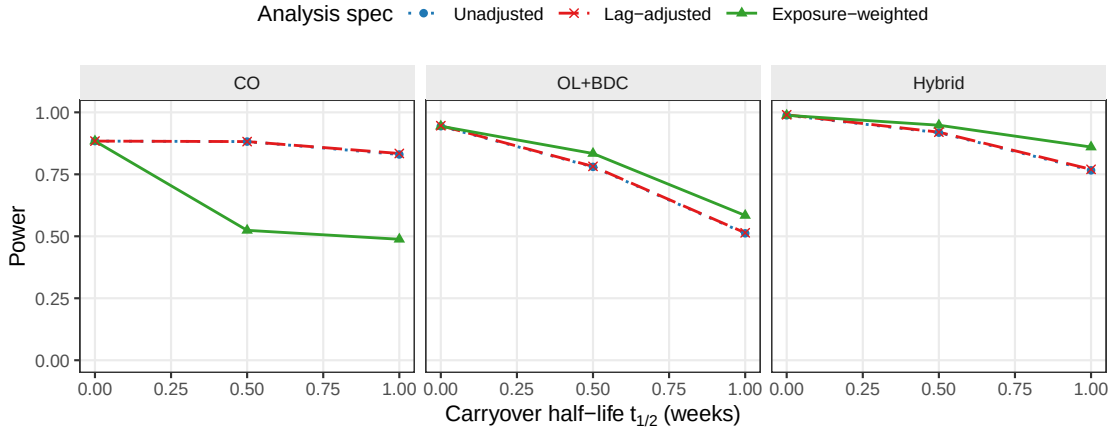


Figure 5: Power versus carryover half-life under the exponential DGP, by analysis specification and trial design.

Two features of Figure 5 carry the substance of this study. First, the Unadjusted and Lag-adjusted curves lie on top of one another in every panel and at every half-life. At the reference cell they give 0.766 and 0.770 at $t_{1/2} = 1.0$ weeks, and across all 216 cells the mean difference is 0.003, with mean bias, MSE, and coverage agreeing to three decimal places. Since the two share an estimand, are fitted to the same simulated datasets, and differ by exactly one nuisance term, this is as clean a null as the design can produce. The lagged-treatment term contributes nothing on any measure (Section 4.1). Second, the three specifications give nearly identical power at $t_{1/2} = 0$, where there is no carryover to model. They diverge as the half-life increases, with Exposure-weighted separating from the other two. It retains a modest but consistent advantage in the Hybrid and OL+BDC designs from $t_{1/2} = 0.5$ weeks onward. Under the crossover design, though, it is markedly inferior to both alternatives (0.488 against 0.830 for Unadjusted). We think this happens because the within-subject correlation in a crossover trial already carries most of the signal the decaying predictor targets. It gains little there, while its down-weighting of off-drug observations works against it (Section 4).

3.5 Decay-form mis-specification

The heatmap cell at the intersection of the exponential DGP row and the Exposure-weighted column is the matched reference, with power 0.850 ± 0.016 in this figure's own simulation (a dedicated decay-shape run, Section 2.6; the 0.860 reported for the same nominal cell in Table 2 and Figure 5 comes from an independent draw of the same DGP, and the two agree within one

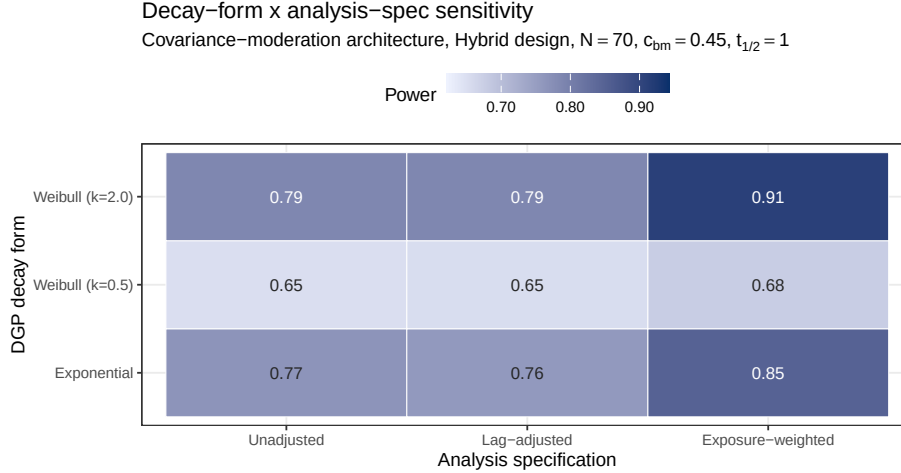


Figure 6: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Hybrid, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms (rows) and specifications (columns). Fill scale is stretched to the observed range, not $[0,1]$, and not comparable to other figures.

MCSE). The Unadjusted and Lag-adjusted columns, invariant to the assumed decay form by construction, are near-identical throughout, as in Figure 5, so variation down those columns is data-generating decay and Monte Carlo noise. The remaining Exposure-weighted rows give the cost of assuming exponential decay when the truth is Weibull. Even at the heaviest-tailed form examined ($k = 0.5$), where the advantage narrows most (power 0.678 against 0.652 and 0.654 for Unadjusted and Lag-adjusted), it does not reverse (Section 4).

3.6 Type I error control and cluster-robust recalibration

Empirical Type I error at the $c_{bm} = 0$ cells of Figure 2 sits uniformly below the nominal 0.05 level for every one of the nine specifications, across every design and N examined there. This is not sampling noise but the conservatism of the model-based test (Section 2.7), from a working covariance that does not match the data-generating one. Because the conservatism is uniform across specifications, it does not disturb the comparison among them, and the cluster-robust recalibration below confirms that a CR2 standard error restores nominal rejection without changing which specification leads. This also establishes that the null result of Section 3.4 is not an artifact of differential test size, since Unadjusted and Lag-adjusted are conservative to the same degree. Table 3 makes this explicit at the null cell, at a higher replicate count (3,000) than the Tier 1 grid affords.

Table 4 confirms this directly. The model-based rejection rate sits below nominal across all five S6 cells and all three specifications, while CR2 restores it to within the Monte Carlo band around 0.05. In the information-rich cells the calibration ratio κ runs from 1.07 to 1.26 under the model-based test, restored to about one under CR2, recovering two to five points of power. The table reports both alternatives against the unadjusted benchmark. The Exposure-weighted advantage is held or slightly widened under CR2 at the reference cell (+0.080 to +0.092), the small- N cell (+0.066 to +0.086), and the half-life mis-specification cell (+0.010

Table 3: Conservatism of the model-based interaction test and its cluster-robust correction, at the null cell ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid design, $N = 70$, 3,000 replicates). κ is the calibration ratio defined in Section 2.7. $\kappa > 1$ indicates a conservative test. All three specifications are conservative under the model-based test, most so Exposure-weighted ($\kappa = 1.10$, Type I = 0.029). The CR2 bias-reduced cluster-robust standard error restores κ to near one and Type I to nominal (0.045 to 0.049) for all three.

Specification	κ model	Type I model	κ CR2	Type I CR2
Unadjusted	1.06	0.036	0.98	0.048
Lag-adjusted	1.05	0.039	0.98	0.049
Exposure-weighted	1.10	0.029	1.00	0.045

Table 4: Cluster-robust recalibration at the reference cell and four stress cells (500 replicates, $c_{bm} = 0.45$). EW is Exposure-weighted, LA is Lag-adjusted, U is Unadjusted. κ is the calibration ratio of Section 2.7, here for Exposure-weighted. A value above one is a conservative test. The two gap pairs give each alternative against the unadjusted benchmark under each inference. CR2 df is the mean Satterthwaite degrees of freedom for the cluster-robust test on Exposure-weighted. Computed with `clubSandwich`.

Cell	κ model	κ CR2	EW model	EW CR2	EW-U model	EW-U CR2	LA-U model	LA-U CR2	CR2 df
Reference, $N = 70$	1.15	0.99	0.829	0.863	+0.080	+0.092	-0.003	+0.002	23
Small $N = 35$	1.13	0.96	0.558	0.590	+0.066	+0.086	-0.002	+0.000	12
High autocorr, $\rho = 0.9$	1.26	0.94	0.954	0.980	+0.052	+0.032	+0.000	-0.004	23
Half-life mis-spec	1.14	1.00	0.759	0.795	+0.010	+0.024	-0.003	+0.002	23
30% MCAR dropout	0.99	0.85	0.148	0.126	+0.010	+0.010	-0.004	+0.010	5

to +0.024). It narrows under high autocorrelation (+0.052 to +0.032, though it still leads), and is small under 30% MCAR dropout (+0.010), which we read as the S3 finding restated, since dropout removes the off-drug information the advantage depends on. The Lag-adjusted gap, by contrast, is within ± 0.010 of zero at every cell under either inference, corroborating the Section 3.4 null under a second procedure and at four stress cells the main grid does not cover. This recalibration was validated only in the Hybrid, $N = 70$ cells, and should not be extended to the crossover design, where Exposure-weighted does not lead in the first place.

3.7 Sensitivity analyses

Nine Tier 2 blocks (S1-S4, S7, S8, S9, S10, S11) perturb one axis at a time around a common reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$), holding the remaining factors fixed. Two exceptions are S9, which changes the trial design itself, and S11, which sets $c_{bm} = 0$. A tenth block, S6, applies the cluster-robust correction of Section 3.6 at that same cell plus four stress cells, and an eleventh, S5, is exploratory and reported only in the archived full-scope drafts. The blocks are reported separately here and are not pooled with Tier 1 or with each other. Four blocks (S1, S4, S10, S11) are reported here only as a one-paragraph summary, with full cell-by-cell detail, tables, and figures in the Supplement; two more (S2, S8) keep their headline finding and a summary paragraph here, with the remainder of their detail likewise moved to the Supplement.

3.7.1 S1: within-factor autocorrelation

All nine specifications gain power monotonically with autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. The Exposure-weighted advantage over Unadjusted and Lag-adjusted, roughly 0.05 to 0.08 in absolute power, is preserved at every level examined. This suggests that autocorrelation strength and specification choice act additively rather than interacting. Lag-adjusted tracks Unadjusted throughout (Supplement, Section S1).

3.7.2 S2: analyst-truth half-life mismatch

Block S2 crosses two true half-lives ($t_{1/2} \in \{0.5, 1.0\}$ weeks) with four analyst-assumed values ($\hat{t}_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks). The penalty for an incorrect assumption is modest throughout (Exposure-weighted stays within about 7 points of its correctly-specified power at every cell but one), with an asymmetry: under-assuming the half-life is safer than over-assuming it. This follows from the limiting-case relationship of Section 2.4, since shrinking $\hat{t}_{1/2}$ drives the predictor toward the binary indicator, which remains serviceable, whereas inflating it drives the predictor toward a constant, at which the on-off contrast and the estimand itself degenerate. Full cell-by-cell detail, figures, and the nine-specification extension are in the Supplement, Section S2.

3.7.3 S3: dropout

S3: Sensitivity to dropout

MCAR and MAR (severity-biased); reference cell as above

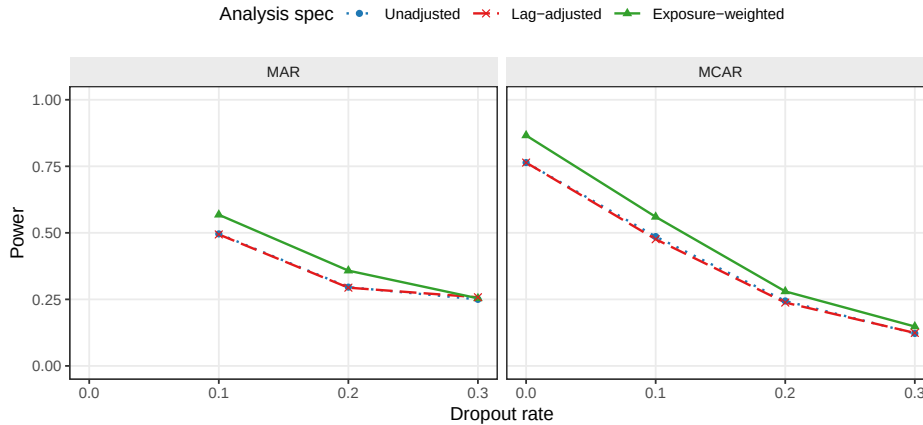


Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms.

All three specifications lose power monotonically as dropout increases. At the MCAR baseline ($N = 70$, no dropout), Exposure-weighted gives 0.866 ± 0.015 against 0.764 ± 0.019 for the other two. At 30% MCAR dropout, all three collapse to 0.122 to 0.148. The Exposure-weighted advantage narrows steadily as dropout worsens, from about 0.10 absolute at zero dropout to 0.07 at 10%, 0.04 at 20%, and 0.03 at 30% under MCAR, narrowing further under MAR at the highest rate. This is consistent with the advantage originating in the exposure-weighted contrast from off-drug observations, which contracts as those observations are preferentially

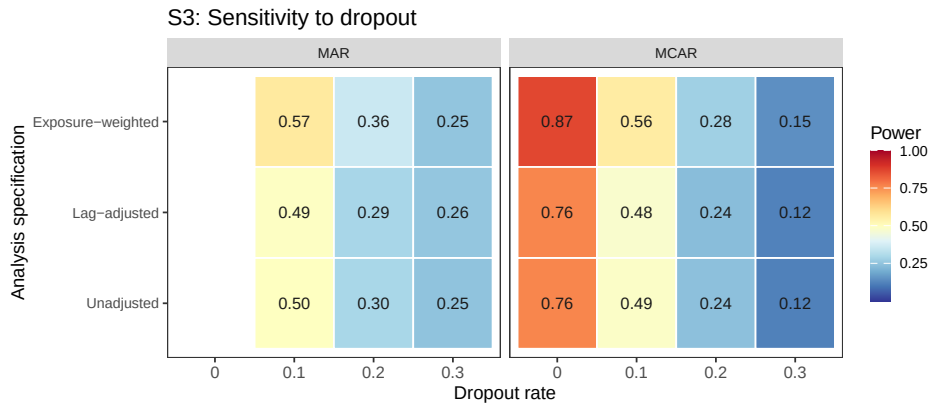


Figure 8: Block S3: power by dropout rate and analysis specification, faceted by dropout mechanism, same data as the line plot above. The MAR arm has no 0% cell (Section 3.7); it shares the MCAR reference cell.

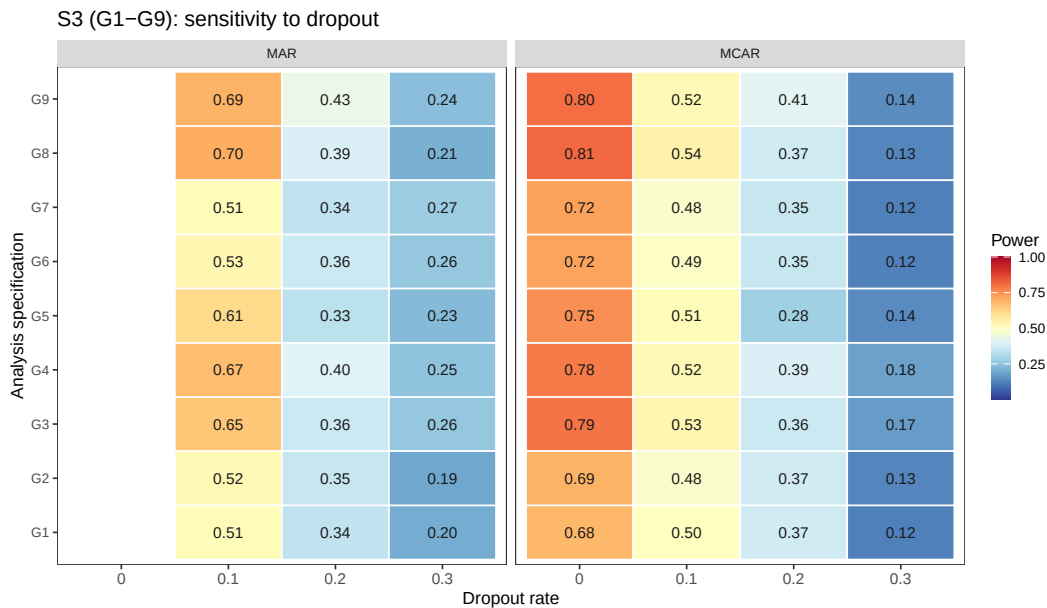


Figure 9: Block S3 extended to all nine G1–G9 specifications (preliminary $n_{sim} = 100$).

Table 5: Block S7: all five G1-G5 specifications at the reference cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$, exponential DGP, $n_{sim} = 500$, all replicates converged). G4 targets bm:Dbc at the AIC-selected half-life (candidate grid 0.25, 0.5, 1.0, 2.0); G5 targets the slope of the patient-level diff bm regression. Neither is the same estimand as bm:Db (G1/G2), so only power is comparable across rows. Mean selected t1/2 is the average AIC-chosen half-life for G4 across replicates, against a true half-life of 1.0.

Specification	Power (MCSE)	Mean selected t1/2
G1: Unadjusted	0.748 (0.019)	–
G2: Lag-adjusted	0.748 (0.019)	–
G3: Exposure-weighted	0.826 (0.017)	–
G4: AIC-selected t1/2	0.832 (0.017)	1.31
G5: Paired-difference	0.804 (0.018)	–

lost. The advantage is preserved at dropout of 20% or below. Lag-adjusted again tracks Unadjusted throughout, the largest discrepancy being 0.010.

3.7.4 S4: biomarker-moderation effect-size curve

The power curve is sigmoidal for all three original specifications across $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$, saturating once $c_{bm} \geq 0.45$. The Exposure-weighted advantage over the unadjusted benchmark is largest at intermediate effect sizes (+0.10 absolute at $c_{bm} = 0.45$), shrinking as all three approach the ceiling, so the advantage matters most for trials targeting a moderate biomarker-treatment interaction (Supplement, Section S4, full curve and nine-specification extension).

3.7.5 S7: alternative-paradigm specifications

An earlier version of this block tested two carryover terms crossed with the biomarker, E5 ($Sx \sim bm + t + D_{it} + bm:D_{it} + L_{it} + bm:L_{it}$) and E6 ($Sx \sim bm + t + D_{it} + bm:D_{it} + t_{sd} + bm:t_{sd}$). This was motivated by Section 4.1's observation that a carryover term entering only as a nuisance main effect cannot repair the interaction's attenuation. Both underperformed Unadjusted at every cell examined, evaluated by a joint 2-df Wald test since each adds a second interaction coefficient. We traced this to a structural cause discussed in Section 4.6: this manuscript's DGP has a single true decay parameter, so a second, DGP-unmotivated interaction coefficient mostly adds estimation noise. E5/E6 were replaced by two specifications addressing the same open problem, the analyst's unknown decay half-life, from different angles (Section 2.4): G4 (AIC-selected half-life) and G5 (paired-difference regression). Both are single-coefficient tests, unlike E5/E6.

Table 5 gives a genuinely different result from the retired E5/E6 comparison. Both G4 and G5 beat Unadjusted at this cell, and G4 edges out G3 (Exposure-weighted) as well. G4 leads overall (0.832), G3 follows (0.826), then G5 (0.804, the two-stage paired-difference regression with no repeated-measures model at all), and G1/G2 tie at the back (0.748 each, the already-established Section 3.4 null replicated once more).

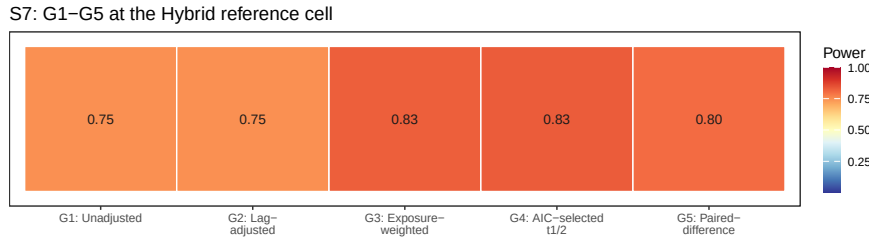


Figure 10: Block S7: power by analysis specification, same figures as the table above.

G4’s AIC selection is imperfect but usefully informative. The mean selected half-life is 1.31 weeks against a true value of 1.0, and the selection distribution concentrates on the true value and its immediate neighbor (88% of replicates choose 1.0 or 2.0 weeks). This is close enough to the truth that G4 slightly exceeds G3’s own oracle-half-life power at this cell.

We see a more notable result in G5. It discards the repeated-measures structure, the correlation model, and any carryover representation entirely, and still beats G1/G2. This echoes Senn, Julious and Araujo’s⁷ finding that simplicity is not obviously costly for N-of-1 treatment-effect estimation. None of these results should be read as general, however. Block S9, next, tests whether the ranking survives a different trial design, and it does not for every specification equally.

3.7.6 S8: architecture sensitivity of the specification ranking

Every result reported to this point, Tier 1 and blocks S1-S7 alike, is restricted to Architecture B (covariance moderation), the manuscript’s stated scope (Section 2.3). Block S8 asks a narrower question than a full architecture comparison, which is the purpose of the companion manuscript on DGP architectures³: does the ranking among analysis specifications established above under Architecture B hold under Architecture A (mean moderation) as well, or is the choice of carryover representation itself architecture-dependent? We refit the five specifications of block S7’s table at the same reference cell under both architectures, at three half-lives ($t_{1/2} \in \{0, 0.5, 1.0\}$).

The S7 result (G4 and G5 both beating G1) turns out to be architecture-specific, joining Exposure-weighted in a pattern that grows with carryover severity rather than switching on at some threshold. Under Architecture B, G4 leads at $t_{1/2} = 1.0$ (0.850), ahead of G5 (0.842), G3 (0.836), and G1/G2 (0.760). Under Architecture A the pattern reverses for G4 and G3 alike: G1/G2 lead (0.972/0.976), G5 trails only slightly (0.964), while G3 (0.946) and G4 (0.950) trail by 2 to 3 points, a real if modest reversal. This matches the archived full-scope drafts’ finding that Exposure-weighted is marginally dominated under mean moderation (Section 4.6). G4 inherits the same sensitivity, since it shares Exposure-weighted’s formula. G5 is comparatively more robust to architecture, never reversing to a large disadvantage either way. Lag-adjusted (G2) remains statistically indistinguishable from Unadjusted (G1) under both architectures at every half-life, extending Section 3.4’s clean null to Architecture A as well. The one specification recommendation this manuscript makes, exposure weighting in the Hybrid and OL+BDC designs (Conclusions), is therefore itself conditional on Architecture B

and grows more important as carryover worsens. The same qualification now extends to **G4**. Full cell-by-cell detail, the table, and both architecture panels are in the Supplement, Section S8.

3.7.7 S9: does the reference-cell ranking survive under CO?

Blocks S7 and S8 found **G4** and **G5** both beat Unadjusted at the Hybrid reference cell under Architecture B, with **G4** slightly exceeding Exposure-weighted (**G3**) itself. Block S9 asks whether that ranking is specific to Hybrid's short, compressed washout, by refitting all five G1-G5 specifications under the CO design (off-drug arm spanning 2.5 to 20 weeks post-discontinuation, Section 2.2) at two true half-lives.

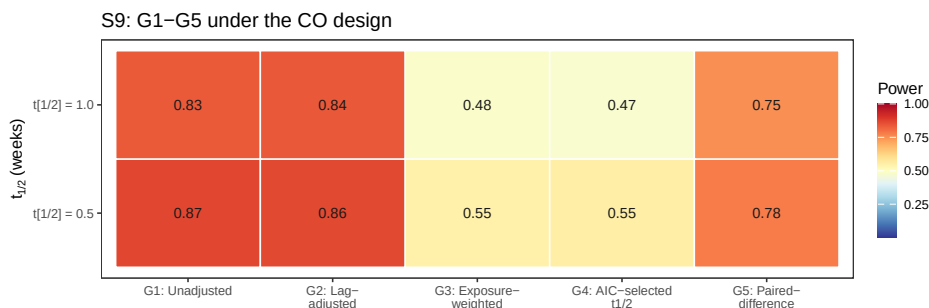


Figure 11: Block S9: power under the CO design at two true half-lives, all five G1-G5 specifications.

It is not, and including **G3** in this block changes the diagnosis reached from **G4** alone. At $t_{1/2} = 1.0$, **G1/G2** lead (0.834/0.838); **G5** trails moderately (0.750); **G3** and **G4** both collapse, to 0.482 and 0.474 respectively, statistically indistinguishable from each other (gap 0.008, well under one MCSE). At $t_{1/2} = 0.5$ the same pattern holds (0.866/0.864 for **G1/G2**, 0.778 for **G5**, 0.546/0.552 for **G3/G4**, again indistinguishable). Because **G3** uses the *true* half-life, not an estimated one, its near-identical collapse shows that **G4**'s failure under CO is inherited almost entirely from Exposure-weighted's own established weakness under this design (Section 3.4: 0.488 versus 0.830 in the Tier 1 grid), not created by the AIC-selection step. The selection step does have a real, separately confirmed pathology under CO: it concentrates overwhelmingly on the largest candidate half-life in the grid (2.0 weeks, chosen in 483 of 500 replicates at $t_{1/2} = 1.0$ and 435 of 500 at $t_{1/2} = 0.5$, regardless of which true half-life generated the data), consistent with a weakly informative likelihood surface across candidates when off-drug occasions are few and widely spaced. But that mis-selection costs **G4** essentially nothing beyond what **G3** already loses under CO with the correct half-life in hand. The dominant story is the continuous decayed predictor's mismatch with CO's design generally (Section 3.4). AIC selection is an identified but practically minor addition to that story here, not the primary cause it appeared to be from **G4** in isolation.

G5's milder reversal has no comparable mechanical explanation. It simply reflects that CO's own within-subject contrast (Section 3.4) already captures most of the signal a between-patient paired-difference regression is built to extract, so discarding the repeated-measures structure costs more there than it does under Hybrid. None of **G3**, **G4**, or **G5**'s Hybrid-cell behavior is

709 design-general, then. The design-specific guidance in Section 4.5 ('What analysis is best in
710 which situations, in practice') should be read accordingly: **G3/G4** should not be used under
711 CO regardless of whether the half-life is assumed or estimated.

712 **3.7.8 S10: does a CR2 correction change the ranking among G1-G9?**

713 Section 3.6 (Block S6) showed the model-based interaction test is mildly conservative, and that
714 a CR2 cluster-robust standard error recovers two to five points of power without disturbing
715 the ranking among **G1**, **G3**, and **G2**. That correction had not been applied to **G4** or **G5**. This
716 block asks whether CR2 helps **G4** and **G1/G2** the way it already helps **G3**, and whether it
717 changes which specification leads at the Hybrid, Architecture B reference cell. We use the
718 compact **G1-G9** naming layer for the resulting nine-way comparison: **G6-G9** cross CR2 with
719 **G1**, **G2**, **G4**, and **G3**. **G5** has no CR2 variant, since its paired-difference regression collapses to
720 one row per patient before fitting, leaving no repeated-measures cluster structure for CR2 to
721 correct.

722 CR2 helps every specification it was applied to: **G1/G2** (Unadjusted and Lag-adjusted, both
723 0.748 model-based) rise to **G6** 0.776 and **G7** 0.764. The ranking among the five base specifica-
724 tions is unchanged by these gains, since **G1/G2** remain well below **G3-G5** regardless. What does
725 change the ranking is **G9**: CR2 lifts **G4** (AIC-selected half-life, 0.832 model-based) to 0.870,
726 a 3.8-point gain, the largest CR2 correction observed anywhere in this manuscript, enough
727 for **G9** to overtake every other specification tested, including **G8** (0.863). This did not come
728 with a Type I error cost (Section S11 of the Supplement checks this directly): every replicate
729 converged under **G9**, and the power gain is at least consistent with CR2 correcting real con-
730 servatism rather than papering over a different problem. Full detail, the reference- cell table,
731 and the heatmap are in the Supplement, Section S10.

732 **3.7.9 S11: is G4's Type I error controlled?**

733 **G4** selects its half-life by AIC and then tests the interaction coefficient from the selected model,
734 a post-selection inference problem: the same data drive both the selection and the test, which
735 can in principle inflate Type I error beyond nominal. Block S11 checks this directly, refitting
736 **G1-G5** and **G9** at the null ($c_{bm} = 0$) at the Hybrid reference cell, with $n_{sim} = 2000$ for tighter
737 precision on a rate expected near 0.05.

738 The concern does not materialize. **G4's** Type I error is 0.038, conservative and in the same
739 range as **G1-G3** (0.034-0.036), not inflated by the selection step. **G9** (**G4**+CR2) is 0.052, indis-
740 tinguishable from nominal 0.05. Read together with Section 3.6's power results, **G9's** status as
741 the best-performing specification (Section 3.7, Block S10) is not an artifact of an uncontrolled
742 test: it wins on power while holding Type I error at nominal, the same standard **G3**+CR2
743 (**G8**) is held to. This does not generalize beyond the single cell tested. Full detail and the
744 heatmap are in the Supplement, Section S11.

3.8 A higher-precision check

Because a number of the Tier 1 gaps are small, we reran a focused 24-cell grid (OL+BDC design, $N \in \{35, 70\}$, true half-life $\in \{0.5, 1.0\}$) at 600 trials per cell, all converging. We compared specifications with a paired McNemar test, considerably more sensitive than an independent-groups comparison since all three are fit to the same trials. Type I error across the twelve null cells fell in $[0.003, 0.053]$, so nominal alpha holds. Paired McNemar tests place Exposure-weighted ahead of the unadjusted benchmark at all four alternative cells, by +2.3 to +13.7 percentage points. The largest gap is at the highest-leverage cell ($N = 70$, true $t_{1/2} = 1.0$ weeks, $p = 6.6 \times 10^{-17}$), with $p \leq 0.015$ at the remaining three, a sharper confirmation than the Tier 1 grid alone provides.

One scope caveat governs the rerun's third arm. It was produced by the package's `lme_analysis()` routine, whose carryover option appends time since discontinuation as a linear main effect rather than the lagged indicator L_{it} used in the Tier 1 grid. It is therefore not the Lag-adjusted specification of Section 2.4 and is not reported as such here. That alternative remains a reasonable specification we have not evaluated (Section 4.6).

4 Discussion

4.1 The lagged-treatment term does no work

The clearest result of this study is a negative one. Adding the classical lagged-treatment indicator to the analysis model changed nothing measurable, anywhere in the grid. Across all 216 cells the mean power difference against the unadjusted benchmark was 0.003 against an MCSE of 0.018. Bias, MSE, and coverage agreed to three decimal places, and the equality held in every sensitivity block and all five cluster-robust stress cells under both inference procedures.

The strength of this conclusion rests on the design rather than the grid size. The two specifications estimate the same coefficient, were fitted to the same simulated datasets under common random numbers, and differ by exactly one term. There is therefore no confounding from a change of estimand, from Monte Carlo variation between arms, or from differential test size (both are conservative to the same degree, Section 3.6). What is left is the term itself, and it contributed nothing.

The mechanism follows from where the term sits. Carryover damages the interaction test on two fronts (Section 1). It attenuates the coefficient, because contaminated occasions are coded as drug-free, and it inflates the residual variance, because those occasions are heterogeneous in a way the model does not represent. The lagged indicator enters only as a nuisance main effect, with no $\text{bm}:L_{it}$ product term, so the tested interaction remains $\text{bm}:D_{it}$ and the attenuation is untouched. It can in principle absorb some of the second effect, but evidently absorbs too little to matter, since it marks a single occasion per randomization path and is constant thereafter, unable to describe a graded decline across a two- to four-occasion washout.

A more capable, elapsed-time-indexed nuisance term might succeed where the crude lagged indicator fails. We therefore ran a targeted check on the alternative the reference implementa-

tion provides, $Sx \sim bm + t + Db + bm:Db + tsd$, called the washout-adjusted specification (E4), across the three designs at true half-lives of 0.5 and 1.0 weeks ($N = 70$, $c_{bm} = 0.45$, exponential DGP, 500 replicates). It does slightly better, and the improvement is instructive rather than material. Type I error is at nominal for both (mean 0.050 versus 0.045). Power exceeds the benchmark in all six cells by a mean of 1.3 points (sign test $p = 0.031$), reaching significance under crossover at both half-lives (+1.8, $p = 0.008$; +2.2, $p = 0.003$) and OL+BDC at the longer one (+2.2, $p = 0.015$), but not Hybrid (+1.0, $p = 0.27$). The mechanism is exactly the one Section 2.4 predicts. Point estimates are unchanged, so nothing about the attenuation is repaired. The entire effect runs through the standard error, which only a long enough washout gives it something to describe. The crossover schedule places off-drug occasions 2.5 to 10 weeks after discontinuation, against one to two weeks for Hybrid. The practical ceiling on what any carryover nuisance term can deliver is therefore about two percentage points, and only in long-washout designs, against the nine to eleven points exposure weighting delivers where it works. This check covers only the exponential DGP at a single N and effect size, so it is a probe rather than a fourth arm. It shares every specification's structural limitation of entering carryover as a main effect only (Section 4.6). For practitioners the implication is direct. A carryover nuisance term of either form should not be expected to recover meaningful power lost to carryover, and reporting one does not constitute an effective adjustment for it.

4.2 The exposure-weighted advantage, and where it fails

The exposure-weighted predictor does help, but conditionally. At its best cell (Hybrid, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$) it exceeds the unadjusted benchmark by about 10 points of power (0.860 versus 0.766, roughly five MCSEs and so not attributable to chance), confirmed at all four cells of the Section 3.8 high-precision rerun. Under the crossover design it performs considerably worse (0.488 versus 0.830). We think this happens because the within-subject correlation there already carries most of the signal the decaying predictor was designed to recover, so it gains little while its down-weighting of off-drug observations works against it. The recommendation therefore holds only in the two non-crossover designs, and should not be extended to a crossover trial, where the unadjusted binary indicator is materially better.

That the one specification addressing attenuation is also the only one to gain power, while the one addressing only residual variance gains nothing, suggests attenuation is the mechanism that matters here. We read this as an interpretation consistent with the results, not as a finding the design isolates directly.

4.3 What the robustness checks add

The Section 3.7 sensitivity blocks show that the exposure-weighted advantage, where it exists, is not fragile. It survives incorrect decay-shape assumptions (Figure 6) and an incorrectly assumed half-life, short of a large over-assumption at a short true half-life (Block S2, Supplement Section S2). It also holds its five- to eight-point lead across the autocorrelation range examined (Block S1, Supplement Section S1).

One caveat qualifies the Tier 1 figures throughout. Outside block S2, the analysis-side half-

life is set equal to the data-generating one. This grants Exposure-weighted an oracle no real analyst possesses, so its Tier 1 margins are upper bounds, and block S2 gives the more realistic figure.

4.4 Dropout matters more than the method

The finding with the largest practical consequence concerns missing data, not carryover. When patients drop out (Figure 7), power collapses for all three specifications together, to between 0.12 and 0.25 at 30% dropout, essentially indistinguishable. No specification offers protection, since each can only organize whatever data remain. MAR dropout is somewhat worse than MCAR, because the sickest patients, who carry the most signal, tend to leave first. Retaining patients in the trial yields considerably more power than any choice among these three specifications ever could.

4.5 What analysis is best in which situations, in practice

Collecting the findings above into a decision rule rather than a set of pairwise comparisons:

- **Covariance moderation (Architecture B), Hybrid or OL+BDC design.** Use Exposure-weighted with a CR2 cluster-robust standard error. The advantage holds under decay-shape mis-specification (Section 3.5), under half-life mis-specification short of a large over-assumption (Section 3.7, Block S2), and across the autocorrelation range examined (Block S1), and CR2 restores nominal Type I error without disturbing the ranking (Block S6, validated at $N \in \{35, 70\}$ in the Hybrid design specifically; OL+BDC was not itself stress-tested under CR2). This remains the recommended default: it has been through the full robustness battery G4/G5 have not (next point).
- **G4 (AIC-selected half-life) and G5 (paired-difference) are promising at this same cell, not yet validated defaults.** Both beat Exposure-weighted itself at the Hybrid, Architecture B reference cell (Blocks S7-S8: G4 0.850, G5 0.842, G3 0.836, at $t_{1/2} = 1.0$), and G4 in particular removes the need to assume a half-life at all. Neither has been tested against decay-shape mis-specification, dropout, or under CR2 the way G3 has, so this is grounds to investigate them further, not to displace the established default yet. Adding a CR2 correction to G4 (G9 in the nine-way naming layer) goes further still: at this same cell it outperforms every one of the other eight specifications, including G3+CR2, with no accompanying Type I error inflation (Section 3.7, Blocks S10-S11). This is the single best result anywhere in this manuscript, and also the least validated. It rests on one cell, with none of G4's own robustness gaps yet closed.
- **Crossover (CO) design, any architecture.** Use Unadjusted. Exposure-weighted is markedly worse under CO (Section 3.4), and Block S9 shows G4 and G5 fail here too, G4 catastrophically. G4's AIC half-life selection systematically drifts to the largest candidate in its grid under CO's sparse, widely spaced off-drug sampling (chosen in over 85% of replicates regardless of the true half-life), collapsing its power to roughly half of Unadjusted's. It should specifically be avoided under any design with few, widely spaced off-drug observations. G5 also underperforms Unadjusted under CO, though far more mildly. The classical lagged-treatment remedy adds nothing here either (next point),

including in the one design where its washout is long enough to give it a chance (Section 4.1).

- **Architecture uncertain, or known to be mean moderation (Architecture A).** Use Unadjusted or Lag-adjusted, not Exposure-weighted or G4. Both specifications' advantage under covariance moderation reverses, marginally, under mean moderation (Block S8). G5 is the one specification whose performance does not depend much on which architecture generated the data, small under B, smaller under A, but it has been tested at only one design and reference cell outside the Hybrid comparison (Block S9).
- **Never rely on Lag-adjusted as a carryover remedy.** It is statistically indistinguishable from Unadjusted everywhere tested in this manuscript, Tier 1 and every Tier 2 block alike (Section 4.1). If a reviewer or protocol expects some carryover term to be present, including it costs nothing measurable, but it should not be reported as having adjusted for carryover.
- **Minimize dropout before optimizing specification choice.** It is the single largest lever in the entire grid (Section 4.4), and no specification, including G4/G5, has been shown to offer any protection against it; their behavior under dropout is untested.

4.5.1 Why G4 and G5 succeed at Hybrid but fail under CO

Both new specifications' Hybrid, Architecture B advantage evaporates under the CO design (Block S9), for two different, design-specific reasons rather than a shared structural flaw of the kind that sank E5/E6 (Section 4.6). G4's failure is mechanical. With only a few, widely spaced off-drug occasions, the likelihood surface across candidate half-lives is weakly informative, and AIC selection drifts to the boundary of its candidate grid rather than locating the true value, systematically overestimating the half-life. Section 3.7's S2 block already showed that over-assumption is the more damaging direction of half-life error when the half-life is fixed by the analyst. Under CO, G4 reproduces that same damaging error through the estimation procedure itself, on nearly every replicate.

This is a concrete, testable prediction: G4 should be reliable whenever a design gives several off-drug occasions across a graded range of elapsed time, as Hybrid does despite its short overall washout, and unreliable whenever off-drug sampling is sparse, however long the washout runs in calendar time.

G5's failure has no comparable mechanical account and is milder. The most plausible explanation is that CO's own within-subject contrast (Section 3.4) already carries most of the signal a between-patient paired-difference regression is built to extract, so discarding the repeated-measures and correlation structure costs more under CO than it does under Hybrid, where that structure is doing more work. This was not tested directly and would require comparing G5 against the within-cell decomposition of G3's advantage to confirm.

Two implications follow. First, these results reinforce rather than undercut the earlier E5/E6 diagnosis (Section 4.6). The more fundamental unresolved question is not which design an alternative specification is tested under, but whether the underlying data-generating process rewards that specification's added structure at all, and neither G4 nor G5 was tested against a DGP built to favor them specifically. Second, the practical recommendation is narrower

905 than ‘estimate the half-life’ or ‘use a simpler method’: both **G4** and **G5** are candidates worth
906 validating further within the Hybrid/OL+BDC, Architecture B cell where this manuscript’s
907 other recommendations already concentrate, not general-purpose replacements for Exposure-
908 weighted or Unadjusted.

909 4.6 Scope, related work, and limits

910 This manuscript deliberately narrows the full-scope grid to the covariance-moderation
911 architecture and to the non-linear decay forms, in the interest of a self-contained eval-
912 uation of manageable length. The archived full-scope drafts (`archive/report.Rmd` and
913 `archive/report-slim.Rmd`) show that the exposure-weighted advantage documented here
914 does not carry over to mean moderation, where the biomarker moderates the mean response
915 directly and Exposure-weighted is marginally dominated in every design examined. Readers
916 who need that cross-architecture comparison should consult those drafts rather than extrap-
917 olate from this one. Establishing what the originally published power values (Section 3.3)
918 were computed under remains outstanding work.

919 The crossover tradition surveyed by Jones and Kenward⁴ would lead one to expect a shape-
920 free lagged term to outperform a parametric predictor once the decay shape is mis-specified.
921 Our results do not support that expectation over the range examined, for two reasons. The
922 exposure-weighted penalty for an incorrect half-life is too small to be overtaken (Section 3.7),
923 and more fundamentally the lagged term does not improve on ignoring carryover at all (Section
924 4.1), leaving no advantage for decay-shape mis-specification to erode.

925 One adjacent specification bounds the negative result of Section 4.1. Every carryover term
926 considered there, lagged and washout-adjusted alike, enters as a nuisance main effect with no
927 product against the biomarker. That finding is therefore that a carryover *main effect* does not
928 recover the attenuated signal, not that no shape-free specification can. The natural candidate
929 is a carryover term interacted with the biomarker, either $Sx \sim bm + t + Db + bm:Db + L$
930 $+ bm:L$ or its washout-adjusted counterpart with $bm:tsd$, letting the biomarker effect decline
931 across the washout without committing to a rate.

932 An earlier version of Block S7 tested exactly these two specifications and found a clean neg-
933 ative result at every cell examined. We traced this to a structural cause: this manuscript’s
934 DGP has a single true decay parameter, so a second, interacted carryover coefficient adds
935 estimation noise without a corresponding signal to recover. They were replaced by two spec-
936 ifications addressing the same open problem, the analyst’s unknown decay half-life, from
937 different angles: **G4** (AIC-selected half-life) and **G5** (a paired-difference regression with no
938 carryover representation at all).

939 Both beat Unadjusted at the Hybrid, Architecture B reference cell, **G4** even edging out
940 Exposure-weighted itself (Block S7). Neither survives a change of trial design or architecture
941 (Blocks S8-S9). **G4**’s AIC selection fails specifically under sparse off-drug sampling (Section
942 4.5), and **G5**’s advantage appears to depend on the repeated-measures structure it discards be-
943 ing otherwise underused, which is design-dependent. Block S8 shows the Lag-adjusted null of
944 Section 3.4 replicates under Architecture A, so it is not an artifact specific to covariance mod-

945 eration. Exposure-weighted's advantage, and now G4's, are architecture-sensitive, reversing
946 in direction, though not substantially, under Architecture A (Section 3.7, Block S8).

947 Three further N-of-1 analysis approaches sit outside the G1-G9 family and are not evaluated
948 here, because each requires substantial independent methodological development rather than
949 a drop-in specification within our existing pipeline.

950 Tang and Landes¹⁰ propose formula-based t -tests that correct a single-subject paired-difference
951 test's standard error and degrees of freedom for an estimated first-order serial correlation, the
952 natural refinement G5 lacks (G5 discards within-patient correlation entirely). Their tests have
953 no biomarker-moderation term, though, and would need a `diff ~ bm` extension with a serial-
954 correlation-aware variance to enter our comparison.

955 Schmid and Yang¹¹ describe a Bayesian hierarchical alternative to the frequentist mixed-
956 model and sandwich-estimator family spanning G1-G9, with partial pooling of individual-level
957 treatment-effect, trend, and autocorrelation parameters across patients. They note that car-
958 ryover is tractable only in aggregated, multi-individual analyses like ours, but a biomarker-
959 moderation extension of their model is a separate undertaking.

960 Daza¹² frames N-of-1 inference as a counterfactual, time-varying-treatment problem, esti-
961 mated via g-computation or inverse-probability weighting. This is a structurally different
962 paradigm from G1-G9's explicit exponential-decay or lag covariates: carryover there is con-
963 founding to be controlled rather than a predictor to be modeled.

964 None of the three is a candidate for a tenth specification within this manuscript's scope, but
965 each identifies a direction future work on this estimand should consider.

966 Two further limitations bound what we can conclude. The numbers presented here come from
967 a single parameter set calibrated to a prazosin and PTSD trajectory, so absolute power values
968 should be read as illustrative and the ranking, itself conditional on trial design, as the more
969 portable result. And the sensitivity checks vary one factor at a time and can therefore miss
970 genuine interactions between factors, for instance between high autocorrelation and heavy
971 dropout. The one joint check available to us, block S5 in the full-scope manuscript, revealed
972 no such interaction.

973 5 Conclusions

- 974 1. **The lagged-treatment term does nothing.** It never improved on ignoring carryover
975 altogether, on any measure, in any design, at any half-life, and in every sensitivity block,
976 a clean null given the two specifications share an estimand and differ by a single term.
977 The term enters as a nuisance main effect only, leaving the attenuation of the interaction
978 untouched, and should not be reported as an adjustment for carryover.
- 979 2. **Exposure weighting helps, but conditionally.** It achieves the highest power, about
980 10 points over the unadjusted benchmark, only in the Hybrid and OL+BDC designs,
981 and its lead survives mis-specified decay shape and half-life there. Under crossover it is
982 markedly worse than the binary indicator (0.488 against 0.830) and should not be used.

Where used, we recommend a CR2 cluster-robust standard error (Block S6).

3. **An approximate half-life is adequate, but err short.** A fourfold error at a true half-life of one week costs 6.6 points of power, so a rough pharmacokinetic estimate suffices in practice. The error is asymmetric. Under-assuming drives the predictor toward the still-serviceable binary indicator, while over-assuming erodes the advantage to a tie. Outside block S2 the analyst is granted the true half-life, so the Tier 1 margins are upper bounds.
4. **Dropout matters more than the choice of method.** Power falls sharply as dropout increases, roughly halving at 20%, equally across all three specifications. Preventing dropout is a more effective use of design effort than any choice among them.
5. **The interaction test is somewhat conservative, and correctly so.** All three specifications reject slightly less often than nominal under the null. A CR2 cluster-robust standard error restores the correct rate while recovering modest power, and the exposure-weighted advantage is held or widened under CR2 at four of five S6 stress cells (Block S6).
6. **This is a narrowed slice of a larger study.** Mean moderation and linear decay fall outside this manuscript's scope. The archived drafts `archive/report.Rmd` and `archive/report-slim.Rmd` are the authority on those comparisons and should be consulted before generalizing beyond covariance moderation. We also make no claim of numerical agreement with the originally published power values (Section 3.3).
7. **Carryover terms interacted with the biomarker do not help; two alternative-paradigm specifications do, but only at the reference cell.** Crossing a lagged or washout carryover term with the biomarker gave lower power than the unadjusted benchmark at every cell examined (Block S7's original version). We traced this to a structural cause: this manuscript's DGP has a single true decay parameter, so a second, interacted coefficient adds estimation noise without a corresponding signal to recover (item 1's mechanism, restated). They were replaced by two specifications answering the same open question, the analyst's unknown decay half-life, differently. **G4** estimates it by AIC selection; **G5** discards the decay representation, and the mixed model itself, for a two-stage paired-difference regression following Senn, Julious and Araujo⁷. Both beat Exposure-weighted at the Hybrid, Architecture B reference cell (Blocks S7-S8; **G4** 0.850, **G5** 0.842, **G3** 0.836 at $t_{1/2} = 1.0$). Neither result generalizes: Block S8 shows both reverse, like Exposure-weighted itself, under Architecture A, and Block S9 shows both fail under the crossover design, **G5** mildly and **G4** catastrophically. **G4**'s CO failure has a clear mechanism: AIC selection systematically drifts to the boundary of its candidate half-life grid when off-drug sampling is sparse and widely spaced, reproducing the S2 finding that over-assuming the half-life is the more damaging error, now through the estimation procedure itself rather than a fixed wrong assumption. Both specifications should be read as promising candidates for further validation at the one cell type where this manuscript already recommends Exposure-weighted, not as general-purpose replacements for it or for Unadjusted.

8. **Adding CR2 to the AIC-selected specification gives the single best result in this manuscript, at one cell only.** G9 (G4 plus a CR2 cluster-robust standard error) outperforms all eight other specifications at the Hybrid, Architecture B reference cell, including G3+CR2, and its Type I error holds at nominal despite the post-selection risk G4's half-life selection step raises in principle (Section 3.7, Blocks S10-S11). We read this as a promising direction, not a recommendation: G9 has not been tested against decay-shape mis-specification, dropout, a crossover design, or an alternative architecture, all of which item 2's specification (G3+CR2) has already been checked against.

6 Conflict of interest

The authors declare no conflicts of interest.

7 Funding

This work received no specific funding.

8 Data availability statement

The data and code supporting the findings of this study are available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`. Simulation drivers are at `analysis/scripts/carryover-sensitivity/`. Exact paths for this manuscript are listed in the Reproducibility section below.

9 Reproducibility

This manuscript was rendered with R 4.6.0 inside the project's zzcollab Docker container. The production simulation ran under R 4.5.3 (§3.6), with the image hash in `Dockerfile` and the package set in `renv.lock`. The simulation study uses `nlme` for the continuous-time linear-mixed analysis, `parallel` for the 8-core parameter-grid sweep, and `data.table` or `furrr/purrr` in the two simulation collections. No simulation code runs during knitting, which loads pre-computed checkpoints only. Each driver sets a master seed (20260507) and derives per-replicate seeds from it via `parallel::mc.set.seed`.

Data and code availability. All simulation outputs, cell-level summaries, and driver scripts are versioned under `analysis/scripts/carryover-sensitivity/` and `analysis/data/` in the pmsimstats-ng repository. This includes the principal factorial, the S1 to S5 sensitivity blocks, the S6 cluster-robust recalibration, the washout-adjusted (E4) check of Section 4.1, and the high-precision prototype rerun of Section 3.7. It also includes the S7 alternative-paradigm block (`09-run-sensitivity-s7.R`, `output/09-sensitivity-s7.rds`); the S8 architecture-by-half-life block (`10-run-sensitivity-s8.R`, `output/10-sensitivity-s8.rds`); the S9 CO-design block (`14-run-sensitivity-s9.R`, `output/14-sensitivity-s9.rds`); the S10

CR2 comparison block (15-run-sensitivity-s10.R, output/15-sensitivity-s10-g.rds); and the S11 Type I error block (16-run-sensitivity-s11.R, output/16-sensitivity-s11.rds). All five were added 2026-08-18/19 (master seed 20260415, 500 replicates per cell except S11's 2000; Hybrid reference cell except S9, which uses CO, and S11, which sets $c_{bm} = 0$). E5/E6 (biomarker-interacted carryover terms) were tested in an earlier version of S7-S9 and are documented in Section 4.6; they were replaced by G4 (AIC-selected half-life, reusing the candidate-grid logic of `characterize_carryover()`, R/carryover_analysis.R) and G5 (paired-difference regression). The E1/E3/E2, E7, E9 fitters, and the E1cr2/E3cr2/E7cr2 CR2 variants used by S10 (`cr2_extract()`, `fit_spec_s7`, `simulate_cell_s7`), are appended to `simulation-core.R` alongside the Tier 1 `fit_spec/fit_all_specs` pair, which they do not modify. `spec-labels.R` additionally defines the G1-G9 compact display-code layer (`g_code`, `g_labels`) used in S10's table and figure; G8 (G3+CR2, stored as E2cr2) is reused directly from Block S6 rather than rerun, so it does not share common random numbers with the rest of S10's comparison. Figure 4, and the decay-form mismatch panel (Figure 6), retain the original three specifications (G1-G3), since decay shape is the only axis those two figures vary. Both were regenerated from a dedicated 18-cell simulation at a more extreme Weibull shape range ($k = 0.5, 2.0$, replacing an earlier $k = 0.7, 1.5$; $k = 1.0$ is not a separate condition, since it duplicates Exponential exactly) than the shared production grid (02-grid-summary.rds) still uses (23-run-decay-shape-sensitivity.R, 02-decay-shape-sensitivity.rds, $n_{sim} = 500$, Architecture B only). This is a standalone script rather than a change to 01-run-factorial.R's shared grid, since that grid is also consumed by the archived full-scope drafts and other manuscripts. Figures 2 and 3 were later expanded to all nine G1-G9 specifications, from a dedicated 30-cell simulation restricted to the exponential DGP and Architecture B (19-run-tier1-hendrickson-g9.R, 02-grid-summary-hendrickson-g9.rds, preliminary $n_{sim} = 100$). This replaced the original three-specification version of both panels rather than supplementing it, and superseded the separate Type I error checks the manuscript previously reported at Figures 6-9 (the Tier 1 Type I boxplot/heatmap and the S6 G1-G3/G1-G9 Type I dot plot/heatmaps), since the $c_{bm} = 0$ column of Panel A now reports Type I error for all nine specifications directly. Those retired figures' underlying data and scripts (03-render-figures-extra-slim.R, 07-type1-cr2-figure-extra-slim.R, 18-run-sensitivity-s6-g9.R) are retained for archival purposes but no longer referenced by this document. The corresponding Tier 2 heatmaps (S1-S4, S6, S7, S8, S9, S10, S11) are produced by 13-hendrickson-heatmap-tier2.R from 02-sensitivity-summary.rds and the S7/S8/S9/S10/S11 output .rds files (plus diag-s6-cr2.rds for G8 and Table 4); these Tier 2 figures are additional to, not replacements for, the existing S1-S4 line plots and the S7/S8 kable tables. The grid design and Morris ADEMP pre-registration are in simulation-grid-plan.md in the same directory. Superseded wider-scope drafts are retained under archive/, documented in its README.md.

Specification labels. All nine specifications are stored under E-prefixed codes (E1, E2, E3, E7, E9, E1cr2, E2cr2, E3cr2, E7cr2) in every data file and script, and this manuscript displays them uniformly as G1-G9, or by name, at render time only. The mapping is applied by `spec-labels.R`, so the stored values and archived outputs are unaffected: what a reader sees as G3 (Exposure-weighted) in the prose above is stored as E2 in every .rds file this

manuscript reads. Output archived before 2026-07-31 stores the specifications as A1, A2, A3.
The migration is documented in `analysis/report/NOTATION.md`.

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